
What Makes a Latent Fire?

Discovering Causal Dependencies Between Sparse Autoencoder Latents

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Abstract

Circuit discovery typically explains an output: a logit, a probability, a behavioural metric. We ask instead what makes an internal latent fire. For a target sparse autoencoder (SAE) latent, *weighted circuit masking* (WCM) learns which upstream latents belong to its circuit and a scaling coefficient for each, trained so that the circuit alone reproduces the target under zero ablation and under mean ablation over activating and contrast contexts at once. On 15,046 sampled TuringLLM latents, 46% of weighted circuits pass one pre-stated criterion on held-out contexts, most often for shallow and residual-stream targets and least often for attention outputs [*pending: Pythia-70m and GPT-2 small, DAN-40/41*]. The resulting weighted circuits hold 10^2 – 10^3 upstream latents and, where they pass, reproduce their target on held-out contexts, while the same nodes at their natural scale reproduce almost nothing and random circuits given the same coefficient fitting fail. The coefficients therefore locate a redundancy that, for deep targets, costs $\sim 10^4$ latents at natural scale, about 14× the weighted circuit. Node sets are not identifiable: refits that leave faithfulness unchanged share only 40–85% of their nodes, so node sets are reported as equivalence classes. The procedure needs no behavioural task and applies to any latent in an SAE bank.

1 Introduction

Sparse autoencoders (SAEs) decompose a language model’s activations into latents that often track coherent concepts (Cunningham et al., 2024; Bricken et al., 2023; Gao et al., 2024; Templeton et al., 2024), a working answer to superposition (Elhage et al., 2022). A latent says what the model represents; it does not say how the model computes it. That question belongs to circuit discovery, whose methods predominantly target an output: a logit, a probability, a behavioural metric. This holds whether the nodes are architectural components (Conmy et al., 2023; Syed et al., 2023) or sparse features (Marks et al., 2025; Ameisen et al., 2025), and whether the circuit is found by search, attribution, or learned optimisation (Bhaskar et al., 2024; Kharlapenko et al., 2025). Output targets bound the explanation to variables we can name as behaviours, while a model computes many internal variables that never surface as any single output.

Latent targets have been approached by attribution: methods that trace upstream of a chosen SAE latent on prompts where it fires and threshold the scores (Ge et al., 2024; Golimblevskaia et al., 2025). The optimised cell of this grid is, to our knowledge, empty: a circuit of SAE latents *optimised* for a single target latent’s activation and tested by intervening on that latent. We call this *latent-targeted circuit discovery*, and work in that cell (Figure 1).

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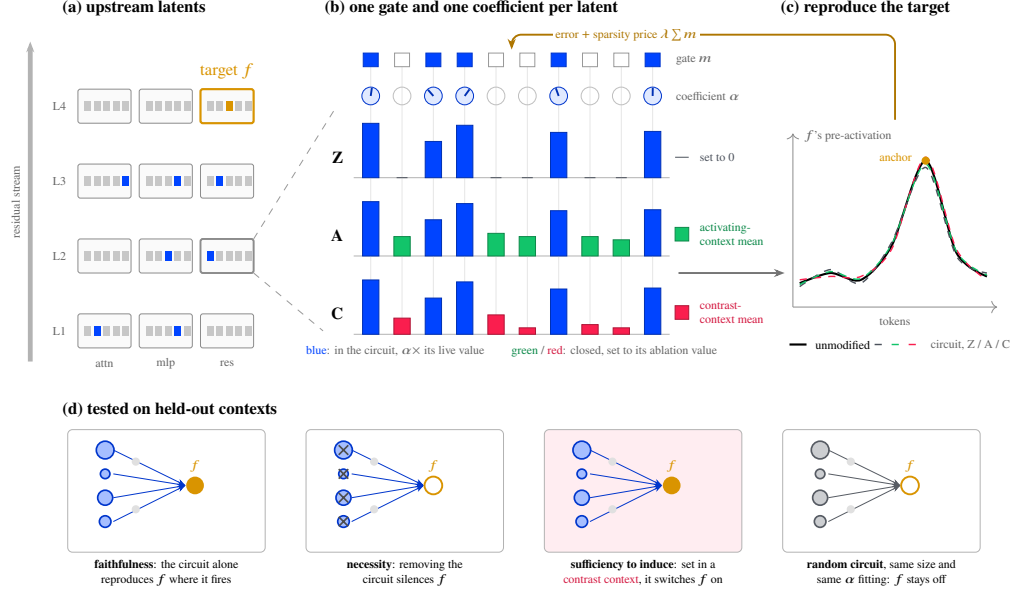


Figure 1: How weighted circuit masking works. **(a)** For a target latent f , every latent at every upstream SAE site is a candidate node. **(b)** Each candidate gets a gate m (kept or closed) and a scaling coefficient α : a kept latent contributes α times its live value, and a closed one is replaced by its ablation value. One set of gates and coefficients is used under three ablation methods at once. **(c)** They are trained so that, under each ablation method, the circuit reproduces f 's pre-activation at its anchor positions, with a price on every open gate. **(d)** The resulting weighted circuit, typically 10^2 – 10^3 latents, is tested on held-out contexts by faithfulness, necessity, and sufficiency to induce, against random circuits of the same size given the same coefficient fitting.

Our method, *weighted circuit masking* (WCM), learns for every upstream latent a gate, whether it belongs to the circuit, and a scaling coefficient, how strongly it acts. Both are trained so that the circuit alone reproduces the target's pre-activation, jointly under zero ablation and under mean ablation over activating and contrast contexts. The result is a *weighted circuit for a target latent*: a node set with one coefficient per node, reusable across the target's contexts. We score it on held-out contexts by three tests: *faithfulness* (the circuit alone reproduces the target where it fires), *necessity* (removing the circuit removes the target), and *sufficiency to induce* (setting the circuit's values switches the target on in contrast contexts), each against random circuits given the identical coefficient-fitting budget.

The main result is how often such circuits exist and how small they are. On 15,046 sampled TuringLLM latents, 46% of weighted circuits pass (65% of residual-stream, 52% of MLP, and 14% of attention-output targets), with a median of 494 upstream latents and 91% within 10^2 – 10^3 [*pending: Pythia-70m and GPT-2 small, DAN-40/41*]. The coefficients are what make the circuits small: the same nodes at their natural scale reproduce almost nothing, faithfulness at natural scale needs $\sim 10^4$ latents for deep targets, about $14\times$ more, and random circuits of the same size with fitted coefficients fail. No behavioural task enters at any point, so the procedure applies to any latent in an SAE bank. We contribute:

- **Weighted circuit masking** for latent-targeted circuit discovery (Section 4.3): a learned node set with fitted scaling coefficients, trained jointly under three ablation methods, with a pre-activation objective that keeps a gradient where Top-K censors the target.
- **Weighted circuits at scale** (Section 5.1): 15,046 sampled TuringLLM targets, with circuits of 10^2 – 10^3 upstream latents tested for faithfulness, necessity, and sufficiency to induce on held-out contexts, 46% of which pass one pre-stated criterion; [*pending: Pythia-70m and GPT-2 small, DAN-40/41*].

- **An evaluation protocol for latent targets** (Section 4.4 and Appendix B): every score names its ablation method and reads the target one way, random-circuit baselines receive the same coefficient fitting, and each method is shown to win the scores it was trained on.
- **Measurements of the circuits themselves** (Sections 5.2, 5.4 and 5.5): the scaling coefficients index the faithfulness cost of natural scale; node sets are identifiable only up to a large equivalence class; and output-targeted and latent-targeted circuits fail each other’s objectives.

We release the pipeline, discovery methods, and analysis code at github.com/DanielJamesDavies/turing-explorer-circuit-discovery.

2 Related Work

Circuit discovery. Manual analysis of model components (Olah et al., 2020; Wang et al., 2023) gave way to automated search (Conmy et al., 2023) and to attribution patching and its integrated-gradient refinements (Nanda, 2023; Syed et al., 2023; Kramár et al., 2024; Hanna et al., 2024). Sparse feature circuits (Marks et al., 2025) moved the nodes to SAE features with error nodes for the reconstruction residual, and attribution graphs (Ameisen et al., 2025; Lindsey et al., 2025) build per-prompt graphs over transcoder features at production scale. Their objectives are anchored at an output, and circuits for one task can differ across its prompts (Makou et al., 2026).

Circuits for internal features. Internal targets are not new. Hierarchical attribution extracts the subgraph behind a chosen SAE feature (Ge et al., 2024), dictionary features have been traced patch-free in Othello-GPT (He et al., 2024), transcoder features are reverse-engineered from their input circuits (Dunefsky et al., 2024), latent attribution debugs a chosen latent (Dupre la Tour and Mossing, 2025), CircuitLens explains a feature’s activation from upstream interactions (Golimblevskaia et al., 2025), and attribution graphs have been extended to concept directions read by linear probes (Palit et al., 2026). These methods attribute and threshold; circuit probing is closer in spirit, optimising a mask over model weights for a labelled intermediate variable (Lepori et al., 2023).

Masks and scalars. Edge Pruning casts discovery as learning a continuous mask (Bhaskar et al., 2024), on L_0 -gate and continuous-sparsification machinery (Louizos et al., 2018; Savarese et al., 2020), and binary masks over SAE latents have been learned by continuous sparsification for behavioural objectives (Caples et al., 2025; Kharlapenko et al., 2025); with its coefficients fixed at 1 and its objective moved to a latent, this is our unweighted circuit masking. Learned scalars appear with other jobs: constants for *removed* components (Li and Janson, 2024), steering multipliers (Stoehr et al., 2024), regression weights on feature-to-feature edges (Yin et al., 2026), and hand-set rescaling of circuit activations (Gao et al., 2025).

Evaluating circuits. What removing a latent means is an experimental choice that can decide the result (Li and Janson, 2024; Miller et al., 2024; Mueller, 2024); patching in clean values (denoising) and removing them (noising) answer different questions (Heimersheim and Nanda, 2024), and our sufficiency to induce is a denoising-style test. Random circuits as a null for faithfulness follow the hypothesis tests of Shi et al. (2024), and MIB standardises circuit evaluation for output targets (Mueller et al., 2025).

What this paper adds is the conjunction: circuits of SAE latents *optimised* for a single target latent with *fitted* scaling coefficients, reusable across the target’s contexts, tested for faithfulness, necessity, and sufficiency to induce on held-out contexts against random circuits given the same coefficient fitting, for $\sim 15k$ sampled latents per model. Position-aware circuits, contrast contexts, and structure among SAE latents are discussed in Appendix I.

3 Setting

The primary model is TuringLLM-1.0-254M, a 12-layer decoder-only transformer trained from scratch on a documented synthetic corpus of ~ 2 billion tokens generated by Phi-3 Mini (Abdin et al., 2024); a model whose training data is known is a more tractable subject than one trained on an opaque crawl (Appendix A). A bank of 36 Top-K sparse autoencoders, one per site (attention output, MLP output, and residual stream at each layer), maps each activation onto 40,960 latents with $K = 128$ active per position (Appendix A.4). The same protocol is applied to Pythia-70m and GPT-2 small with public Top-K dictionaries (Section 5.1). As a running example we follow a layer-3 residual-stream latent of TuringLLM that fires on *temperature as a thermodynamic state variable*: strongly on passages about heat, equilibria, and kinetics, and not at all on most text.

4 Method

For a latent f and a sequence–position pair (x, t) , write $a_f(x, t) \geq 0$ for its activation (after Top-K) and $\hat{a}_f(x, t)$ for its pre-activation (before it); pseudocode is in Appendix K.

4.1 Problem

Targets and contexts. Fix a target latent f and let U_f be the latents at every SAE site that causally precedes f ’s site. Each target has *activating contexts* from two stores, its 64 strongest-activating sequences and 64 drawn from the middle of its activation distribution, each with an anchor at the target’s peak (Appendix L). It trains on 32 strongest and 16 mid-band contexts and is scored on 16 held-out strongest contexts (primary) and 16 held-out mid-band contexts (reported); the split is stratified by activation and always trains on the two strongest (Appendix B). It also has 64 *contrast contexts*: corpus sequences ranked by their highest cosine similarity to any of its activating contexts in pooled final-layer representation space, excluding those contexts, and kept only if the target stays out of its site’s Top-K on them, so that they are close to the target’s contexts but verified silent (Appendix L). For the running example they are passages semantically adjacent to thermodynamics [*pending: running-example contrast contexts under the v1 selection*]. Targets are sampled at random, stratified across sites, with no prioritisation heuristic.

Circuit-only execution. A *weighted circuit* is a node set $C \subseteq U_f$ with a scaling coefficient $\alpha_{s,i} > 0$ for each node. Under an ablation method π , the forward pass is modified at every upstream site s : the site’s Top-K code c_s , computed from the (already modified) stream, is replaced by

$$\hat{c}_{s,i} = \begin{cases} \alpha_{s,i} c_{s,i} & (s, i) \in C \\ v_{s,i}^\pi & \text{otherwise,} \end{cases}$$

and the stream continues as $x + W_{\text{dec}}^s(\hat{c}_s - c_s)$, so SAE errors pass through unchanged and every downstream site, including the target’s, reads the modified stream. Nodes are *recomputed*, not clamped: a node’s value at each position is whatever the modified stream gives it. The ablation value v^π is zero (Z), the latent’s mean over activating contexts (A), or its mean over contrast contexts (C), optionally sparsity-preserving, with the per-position Top-K re-imposed. An unweighted circuit fixes $\alpha = 1$. Writing a_C for the target’s mean activation at the anchors under this intervention, $a_\emptyset$ for the empty circuit, and a_{pos} for the unmodified model, *faithfulness* is

$$\phi_\pi^{\text{free}}(C) = \frac{a_C - a_\emptyset}{a_{\text{pos}} - a_\emptyset},$$

and *clamped faithfulness* $\phi_\pi^{\text{pin}}(C)$ is the same ratio with nodes clamped to their clean-run values.

The problem. Faithfulness depends on the ablation method (Miller et al., 2024; Mueller, 2024), and a node set trained under one method can be exactly infeasible under another (Section 4.3), so one weighted circuit is asked to serve all three:

$$n_\varepsilon(f) = \min\{|C| : \exists \alpha \text{ with } \phi_\pi^{\text{free}}(C, \alpha) \geq 1 - \varepsilon \text{ for every } \pi \in \{Z, C, A\}\}. \quad (\text{P})$$

WCM solves a differentiable relaxation of (P). With $\alpha = 1$ the smallest faithful node sets are several times larger, and attribution at natural scale needs one to two orders of magnitude more (Section 5.2); (P) asks for a minimiser: its near-optimal solutions form a large, non-nested equivalence class (Section 5.5), so only class-invariant quantities are treated as properties of the target.

4.2 Attribution Baselines

WCM is compared against rank-and-cut attribution. Every upstream latent is scored by gradient-times-activation: a *contrastive* objective on contrast contexts, taken through the target’s pre-activation because Top-K censors its activation to zero there; an *ablation* objective on activating contexts, where the activation is live; and their fusion. Each runs under four linearisation modes; *path attribution* is the integrated-gradients estimator of sparse feature circuits (Marks et al., 2025). Scores are cut per position and the node set is the union over the target’s causal prefix, since a Top-K code is sparse per position while the target reads every position; the 128-target grid of Appendix P.1 predates this construction and uses per-site top- K caps instead (Appendix K). Objectives, modes, and the node-set construction are in Appendix C.

4.3 Weighted Circuit Masking

Rather than thresholding attribution scores, WCM learns the circuit directly, adopting the optimisation framing and gate machinery of learned-mask circuit discovery (Bhaskar et al., 2024; Kharlapenko et al., 2025) and moving its objective from an output to a single latent. For every upstream site s it attaches gate parameters $\theta_s \in \mathbb{R}^{d_{\text{sae}}}$ with gates $m_s = \sigma(\theta_s)$, and coefficient parameters ψ_s with $\alpha_s = \text{softplus}(\psi_s)$. A latent is kept at α times its live value when its gate is open and sent to its ablation value when closed:

$$x \leftarrow x + W_{\text{dec}}^s [m_s \odot \alpha_s \odot c^s + (1 - m_s) \odot c_{\text{abl}}^s - c^s],$$

where c^s is the site’s dense Top-K code and c_{abl}^s its ablation value; editing only the decoded difference leaves the SAE error untouched, as in evaluation.

Objective. The circuit is trained to *reproduce* the target’s pre-activation at the anchors of the training contexts, by mean squared error against its value on the unmodified stream. The pre-activation is forced by Top-K: the activation is censored at zero, so a loss on it carries no gradient exactly where the mask must learn. Reproducing the pre-activation does not, however, keep the target in its site’s Top-K: the loss constrains the site’s stream along one direction of d_{model} , and a circuit can restore that projection while other latents at the site rise past the target. A *rank-keep* term therefore asks the target to hold its natural rank: with r the target’s rank among its site’s other latents at an anchor in the unmodified model, and $\tau_\pi^{(r)}$ the r -th largest of their pre-activations under the circuit,

$$\mathcal{R}_\pi = \text{mean}\left[\text{relu}(\tau_\pi^{(r)} - \hat{a}_f)^2\right],$$

which is zero whenever the target is at or above its natural rank, so it never pulls a circuit towards a state the unmodified model does not have.

Training under three ablation methods. A mask trained under one ablation method learns whatever that method rewards. Trained only under mean ablation over contrast contexts, it reaches high faithfulness under that ablation while its zero-ablation faithfulness is 0.0 at depth: it learns

the difference from the contrast-context mean, never a set that suffices alone. Trained only under zero ablation, it suffices alone but is never asked what separates firing from a near-identical context that does not fire. WCM therefore scores one mask under all three ablation methods at once, with gradients accumulating into the same parameters:

$$\mathcal{L} = \frac{\sum_{\pi} \gamma_{\pi} (\mathcal{L}_{\pi} + \gamma_{\mathcal{R}} \mathcal{R}_{\pi})}{\nu} + \lambda \sum_s \sum_i \sigma(\theta_{s,i}), \quad \nu = \text{mean}(\text{target}^2),$$

summed over $\pi \in \{\mathcal{Z}, \mathcal{C}, \mathcal{A}\}$, with $\gamma_{\mathcal{Z}} = 1$, $\gamma_{\mathcal{C}} = \gamma_{\mathcal{A}} = 0.25$ at every layer, and $\gamma_{\mathcal{R}} = 3 \times 10^{-3}$; the ablation means are computed on the training contexts only. All terms measure the same quantity on the same target, so one shared, bounded normaliser suffices, and the penalty is a flat per-latent price, which outperformed every gradient-informed site weighting we tried (Appendix O). In calibration the $\mathcal{A}$ term roughly halves the smallest passing node set on layer-2 targets and raises the share of layer-9 targets with any passing circuit. The ablation method matters more than one might expect: on four layer-2 targets, node sets trained under different ablation methods at matched size share fewer latents (Jaccard 0.13) than node sets trained for different *objectives* under the same method (0.54).

Scaling coefficients. Coefficients start at $\alpha = 1$, so training starts exactly at unweighted behaviour; *unweighted circuit masking* keeps them there. One guard is required: with coefficients free, the optimiser can hold a gate just below the keep threshold while inflating its coefficient, emptying the node set while keeping function intact, so latents with closed gates are charged $\lambda (1 - m) |\alpha - 1|$. The zero-ablation term anchors the coefficients: trained without it, they calibrate against the mean ablation’s background and overshoot by one to three orders of magnitude once that background is removed (Appendix E).

Binarisation. Evaluation scores the hard node set $\{i : m_i > \frac{1}{2}\}$, so the gates are annealed during training, $m = \sigma(\theta/T)$ with T decaying geometrically from 1 to 0.05 (continuous sparsification; Savarese et al., 2020), and the training forward converges to the binary gates that evaluation scores. Training runs 400 AdamW steps; all settings are in Appendix B, and the binarisation and size-calibration studies in Appendix O.

4.4 Evaluation

Table 1 names the evaluation family; every reported number carries its row, is read on the 16 held-out activating contexts (or held-out contrast contexts), and reads the target’s activation in every term of its ratio (Appendix B).

Intervention scores. Let A^+ and A^- be the target’s mean activation on activating and contrast contexts. **Sufficiency to induce**, $\phi_{\text{ind}} = (A_{\text{int}}^- - A^-)/(A^+ - A^-)$, sets the circuit’s nodes to their activating-context means in contrast contexts: near 1, the circuit switches the target on. **Necessity** is one minus the completeness score of Marks et al. (2025): with $a_{M \setminus C}$ the target’s activation on activating contexts when the circuit’s nodes are ablated per π and every other latent is recomputed,

$$\phi_{\text{sup}}(C; \pi) = \frac{a_{\text{pos}} - a_{M \setminus C}}{a_{\text{pos}} - a_{\emptyset}}.$$

Both scores intervene on the circuit as one set. All scores are unclipped ratios: success means landing near 1, and a score far above 1 is overshoot, not quality.

Ablation methods. The choice of ablation is load-bearing (Li and Janson, 2024; Miller et al., 2024). Zero ablation needs no reference distribution. Mean ablation over activating contexts leaks target signal into the ablation value increasingly with depth ($a_{\emptyset}$ up to 30% of a_{pos}); mean ablation over contrast contexts does not. Dense mean ablation puts the model on streams a Top-K SAE can never emit, so the sparsity-preserving variant re-imposes the per-position Top-K after the means are applied (Appendix M.2).

Score	Contexts	Circuit latents	Other latents
ϕ_{ind}	contrast	set	untouched
ϕ_{sup}	activating	ablated per π	recomputed
ϕ_{pin}	activating	clamped to clean values	ablated per π
$\phi_{\text{Z}}^{\text{free}}$	activating	recomputed (permitted)	zero
$\phi_{\text{A}}^{\text{free}}$	activating	recomputed	activating means, dense
$\phi_{\text{A,sp}}^{\text{free}}$	activating	recomputed	activating means, sp
$\phi_{\text{C,sp}}^{\text{free}}$	activating	recomputed	contrast means, sp
$\phi_{\text{Z},\alpha}^{\text{free}} / \phi_{\text{A},\alpha}^{\text{free}}$	activating	recomputed, $\times \alpha$	zero / activating means
$\phi_{\text{ind}}^{\alpha}$	contrast	set to $\alpha \times$ clean values	untouched

Table 1: The evaluation family. Intervention scores: ϕ_{ind} sufficiency to induce, ϕ_{sup} necessity; “set” means set to activating-context values. Faithfulness ϕ_{π}^{free} measures whether the circuit reproduces the target alone with its nodes recomputed, clamped faithfulness ϕ^{pin} whether it contains the right latents, and each variant names its ablation method π (Z zero, A mean over activating contexts, C mean over contrast contexts; sp sparsity-preserving; Section 4.1). The α rows score weighted circuits with their scaling coefficients applied (Section 5.2). SAE errors are preserved in every row, and every ratio reads the target the same way in all of its terms (Appendix B).

Random-circuit baseline. Fitted coefficients could make any set look faithful, so each random circuit, of the same size and drawn from live latents at the same sites, receives the *same coefficient-fitting budget* on its own nodes before scoring; faithfulness then reflects selection rather than the fitting capacity of the coefficients.

Specificity. Every score above reads the target alone, so a circuit that restores the target by lifting its whole neighbourhood at the site passes them all. We therefore read the full code at the target’s site under the circuit and report, as a measured property rather than a pass criterion: the target’s rank and Top-K membership, the number of latents the circuit lifts above the unmodified model’s Top-K cut relative to the empty circuit, and the faithfulness of *sibling* latents that co-fire with the target, against the target’s own (Appendix H).

Near-threshold targets. A target that only just enters its site’s Top-K in the unmodified model has an ill-conditioned activation read: a circuit can reproduce its pre-activation while small shifts near the cut push it out, so its activation-read scores are near zero whatever the circuit. Such targets, those whose median clean rank at the anchor is at least $K/2$, are identified from the unmodified model alone, before any circuit is fitted; they stay in the headline pass rate and are also reported as a separate stratum on both reads.

Pass criterion. A circuit passes if, on the held-out strongest contexts and the activation read, its faithfulness under all three ablation methods (Z, sparsity-preserving A, and C) lies in $[0.8, 1.5]$ and its necessity is at least 0.9. The band is asymmetric by design: below 0.8 the circuit fails to drive its target, while an overshoot up to 1.5 still drives it, and whether that overshoot is specific to the target is measured for every circuit (above). Each circuit’s largest deviation $\max_{\pi} |\phi_{\pi}^{\text{free}} - 1|$ is reported alongside the binary rule; sufficiency to induce, specificity, and the random-circuit baseline are reported rather than required; targets whose ratio denominator is vacuous are excluded and counted.

5 Experiments and Results

Setup. Results come from three sources. The *production run* fits WCM[Z+C+A] to 15,046 sampled TuringLLM targets under protocol v1 (Appendix B); *pending: Pythia-70m and GPT-2 small, DAN-40/41*. A depth-stratified *panel* of 22 TuringLLM targets carries the matched-size comparisons against other discovery methods; its current numbers predate protocol v1 and are marked as such. A

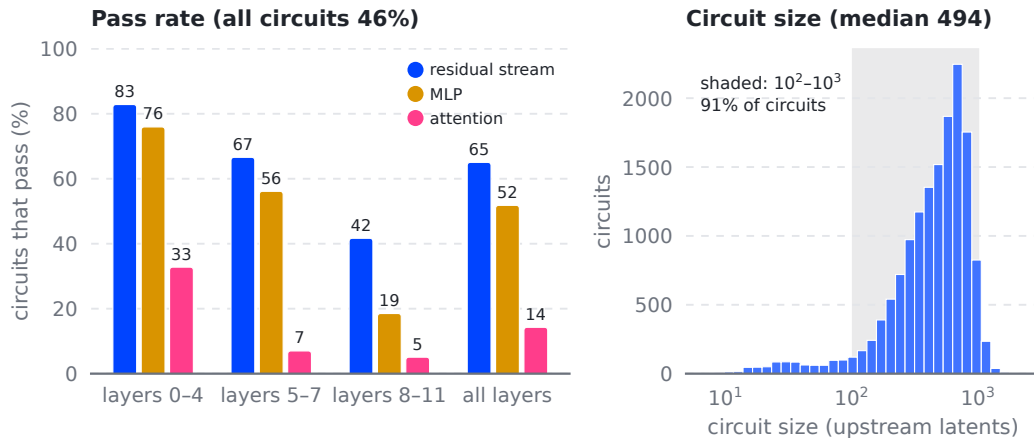


Figure 2: Weighted circuits for 15,004 TuringLLM latents under protocol v1. Left: the share of circuits that pass the pre-stated criterion (faithfulness under Z, A, and C in $[0.8, 1.5]$ and necessity ≥ 0.9 , on held-out contexts), by target depth band and site kind. Right: circuit sizes; 91% lie within 10^2 – 10^3 upstream latents (shaded).

128-target *grid* of an earlier form of the gradient attribution methods grounds the contrast-context and co-activation findings (Appendix P.1).

5.1 Weighted Circuits at Scale

Figure 2 summarises the production run: 15,004 fitted circuits (42 targets had too few activating or silence-checked contrast contexts, and 10 circuits with a vacuous denominator are excluded). Under the pass criterion (Section 4.4), 46% of circuits pass. They hold a median of 494 upstream latents (386 for passing circuits), and 91% lie within 10^2 – 10^3 . The pass rate depends on the target’s site and depth: 65% of residual-stream, 52% of MLP, and 14% of attention-output targets pass, and the rate falls from 85% at layer 1 to 15% at layer 11. Zero ablation holds almost everywhere (median faithfulness 0.94); the failures lie under the two mean ablations, where deep and attention targets fall short (medians 0.64 and 0.48 at layer 11, and 0.28 and 0.24 for attention outputs). Layer 0 departs from the trend (55%): its circuits are faithful (90%), but a third fail necessity, because the token embedding reaches layer-0 targets directly and is never ablated. Near-threshold targets are almost all attention outputs (1,142 targets, 7.6%; 0.8% pass), and the remaining targets pass at 50%. On the held-out mid-band contexts, where targets fire more weakly, 12% pass (median largest deviation 0.77, against 0.32 on the strongest contexts), so circuits fitted mostly to strong contexts transfer only in part to weak ones. Specificity is measured rather than required: 30% of circuits (44% at MLP targets) also restore co-firing sibling latents to at least 80% of the target’s faithfulness (Appendix H). Sufficiency to induce, which no circuit is trained on, is reported separately (Figure 10): its median is 0.98 among passing circuits, 1.19 at residual-stream and 0.20 at attention-output targets. Sizes are at a fixed 400-step budget, at which most circuits are still shrinking (Figure 11).

Beyond TuringLLM. The same protocol is applied unchanged to two public models with public *Top-K* dictionaries: EleutherAI’s $k=16$, 32,768-latent dictionaries for Pythia-70m and OpenAI’s $k=32$, 32,768-latent dictionaries for GPT-2 small, at every layer (sites in Table 4). Only the model and dictionary adapters differ; discovery, the ablation methods, the contrast contexts, and every score run through the same code as on TuringLLM. *[pending: full-run results, DAN-40 and DAN-41, folded into the figure above]*

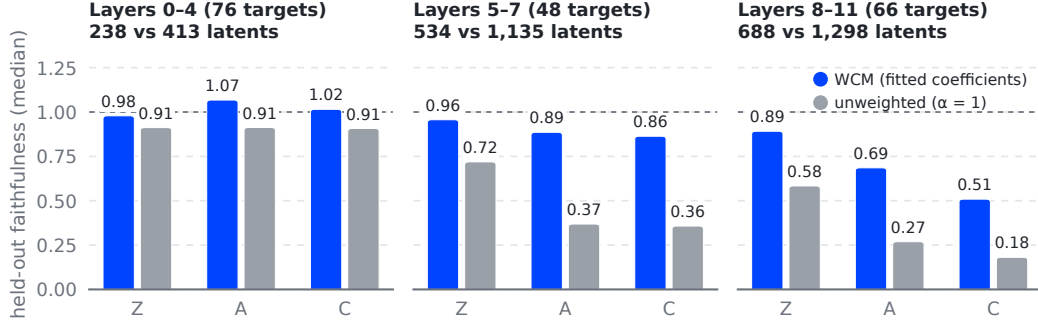


Figure 3: At the same price, fitted coefficients make circuits both smaller and more faithful. Median held-out faithfulness of weighted circuit masking (blue) and unweighted circuit masking ($\alpha = 1$, grey) at one sparsity price, $\lambda = 10^{-3}$ (the production price), under zero ablation (Z) and under mean ablation over activating (A) and over contrast (C) contexts; 1 is perfect (dashed). Panels are target depth bands (190 targets, protocol v1); their titles give the median circuit size, weighted against unweighted. Interquartile ranges and the sweep over prices are in Figure 9.

5.2 Why the Coefficients Matter

Three versions of each circuit separate what the nodes contribute from what the coefficients contribute: the weighted circuit as fitted, the same nodes at natural scale ($\alpha = 1$), and a random circuit of the same size given the same coefficient fitting (Section 4.4). *[pending: the three-way comparison on the production runs, DAN-75/17/20: faithfulness and sufficiency to induce for fitted α , $\alpha = 1$, and random circuits with fitted coefficients, by depth and model]* At natural scale, attribution needs 10^4 – 10^5 latents before the circuit alone reproduces a deep target (Appendix D); the coefficients locate most of that cost in redundancy, many latents each carrying a small share of a signal that fewer, rescaled nodes carry alone. Unweighted circuit masking isolates what the coefficients buy: the same objective and optimiser with α held at 1. At the same sparsity price ($\lambda = 10^{-3}$, the production price), unweighted circuits keep a median 1.85 $\times$ as many latents per target and are still less faithful, and the gap grows with depth, most under the two mean ablations (Figure 3). Matching faithfulness instead of price gives the natural-scale cost: across a sweep of prices, the smallest circuit whose scores all lie within 0.3 of 1 is a median 3.7 $\times$ larger unweighted at layers 0–4, 9.8 $\times$ at 5–7, and 14.1 $\times$ at 8–11, where it holds $\sim 10^4$ latents (Appendix E).

For the running example, the weighted circuit holds 192 nodes and reconstructs the target on held-out contexts, while unweighted circuit masking on the same target needs 577 nodes and still fails under mean ablation (pre-v1 fit; *[pending: refit under protocol v1]*).

Random circuits. Random circuits given the same coefficient fitting failed on every one of 124 draws on the 22-target panel (*[pending: re-scored with the fixed evaluation, DAN-16]*), and none induced the target in contrast contexts, so faithfulness and sufficiency to induce are properties of the selected node set, not of the fitting capacity of the coefficients. *[pending: random-circuit baseline with fitted coefficients at scale on all three models, DAN-17/46, with the fraction of anchors at which a random live latent is nonzero]*

The coefficients carry causal weight. Necessity is unchanged by re-weighting, and the coefficients matter on a test they were never fitted to: setting nodes to α times their clean values induces the target in held-out contrast contexts, most strongly at residual-stream targets (Appendix E; pre-v1 panel, *[pending: sufficiency to induce with and without α after the refit, DAN-16]*).

Arm	n (med)	ϕ_Z^{free}	ϕ_A^{free}	ϕ_{ind}	ϕ_{sup}	pass
path attribution (abl)	591,120	0.99	0.99	1.15	1.00	—
path attribution (ctr)	591,152	0.99	0.99	1.15	1.00	—
restoration (rounds=sites)	560,077	0.99	0.97	1.14	1.00	—
path attr. (abl), top- n	369	1.08	0.00	0.61	1.00	0/21
path attr. (ctr), top- n	369	0.99	0.00	0.56	1.00	0/21
restoration, top- n	369	4.36	0.00	0.66	0.95	0/21
hierarchical attr. (replication)	359	0.63	0.00	0.86	1.00	0/21
unweighted circuit masking	1,216	0.66	0.23	0.75	1.00	3/21
weighted circuit, 100 steps	1,061	0.96	1.00	1.01	1.00	12/21
weighted circuit, 400 steps	369	0.99	1.08	0.89	1.00	17/21

Table 2: The cross-method matrix (pre-v1: fitted on 36 contexts and scored before the evaluation fixes of Appendix B; to be re-scored after the refit): medians over 22 depth-stratified targets, every score on held-out contexts; weighted-circuit rows apply their scaling coefficients (the α rows of Table 1). “pass” counts targets with faithfulness under both zero and mean ablation in $[0.8, 1.25]$ and $\phi_{\text{sup}} > 0.9$, a provisional rule that predates the pass criterion of Section 4.4; one target’s mean-ablation denominator is vacuous and excluded. Top- n rows truncate each attribution ranking to the target’s weighted-circuit size; the hierarchical-attribution row replicates Ge et al. (2024) on our sites with its threshold bisected to the same size.

5.3 Method Comparison at Matched Size

Table 2 compares the discovery families on the same 22 depth-stratified targets, the same held-out contexts, and the same ablation methods: the attribution baselines (Section 4.2) at their natural size and truncated to the weighted circuit’s size, a replication of the hierarchical attribution of Ge et al. (2024), and unweighted and weighted circuit masking. *[pending: table re-scored after the protocol-v1 refit, DAN-16]*

Three results structure the comparison. First, at natural size every attribution arm reconstructs the target with its full per-position union, 10^5 – 10^6 nodes (the smallest faithful prefix of the same ranking is the 10^4 – 10^5 of Section 5.2); at that scale the arms’ node sets largely coincide, so method identity matters only under compactness pressure. Second, truncated to the weighted circuits’ size, every attribution arm fails, including the replication of Ge et al. (2024), whose threshold is bisected per target to matched size; unweighted circuit masking rarely passes; and the weighted circuit passes on most targets. Compact faithfulness is reached only by optimising the node set *and* its coefficients together; the truncations are naive, but no re-ranking closes a gap from 0.00 to ~ 1 under mean ablation at matched size. Third, a method wins the scores whose ablation method it was trained or ranked under, which we call the *home-turf effect* and observed repeatedly across this programme, and under the prior work’s own first-order evaluation the ranking inverts (Appendix E); rankings are therefore a matrix of discovery method against ablation method, never a single column.

5.4 Secondary Findings

Contrast contexts are a controlled variable. Semantic closeness and clean causal contrast are not the same property: for the contrastive attribution baseline, random contrast contexts beat the store’s unchecked close ones on sufficiency to induce, while WCM circuits refitted under close, random, or distant silence-checked contexts move about as far as an identical refit does (Appendix G); *[pending: train-source $\times$ evaluation-source matrix at 288 targets, DAN-64]*.

Discovered dependencies are not co-activation. Across 7,525 accepted attribution circuits, a circuit’s nodes overlap its target’s top co-activation partners by $\sim 3.6\%$, and correlational proposals fail the intervention gates outright; even granted fitted coefficients, a co-activation selection passes single scores but never the joint suite (Appendix F).

Output-targeted circuits do not explain latents, and vice versa. A replication of sparse feature circuits (Marks et al., 2025) on the same model, SAE bank, and targets, with the same attribution estimator, scores $\phi_{\text{ind}} = \phi_{\text{sup}} = 0.000$ on every target latent even at its most permissive threshold, and our circuits score near zero on logit faithfulness (Appendix L.1): “which latents support this output?” and “what makes this latent fire?” are answered by circuits that fail each other’s objective. The output-targeted arm ran at its ceiling threshold without a held-out split and before protocol v1, so this is a scope result, not a ranking; *[pending: re-run under protocol v1, DAN-48]*.

A feature relayed through the residual stream. Wiring the weighted circuit of a layer-11 MLP latent that fires on “pivotal” with direct-effect edges (Appendix N.4) shows the residual stream carrying one feature forward: every upstream residual dictionary’s best detector of the subword “iv” is a node, and edges shared by all three ablation methods link each layer’s copy to the next. The copies are one feature: consecutive decoder directions are nearest or second-nearest neighbours among 40,960 latents (cosine 0.38–0.68) and co-fire on random text ($r = 0.87$ –0.95).

5.5 Node Sets Are Non-Identifiable; Function Is Not

The masking engine is bit-deterministic at fixed settings (two identical runs agree to six decimal places with *zero* node flips), so tiny perturbations are controlled experiments on the objective, and every one we tried gives the same verdict (Appendix J). Float precision, the sparsity price, the stopping step, the training budget, and the contexts behind the mean-ablation values all leave size and faithfulness unchanged but exchange nodes: two node sets for the same target share only 40–85% of their latents. A discovered node set is one sample from an equivalence class of near-optimal solutions to (P), so we report node sets as equivalence classes and let set-level and functional claims carry the conclusions.

6 Discussion and Limitations

Circuit size is a measurement at a stated scaling convention. At natural scale a latent’s circuit is large for deep targets (Section 5.2), and every one of those latents is individually removable at the noise level while bulk removal craters the target (Appendix D). A latent’s activation, reached through an additive residual stream, is a dense function of many small contributions, so any rule that thresholds individual contributions undercounts it however the threshold is tuned. Fitted coefficients show that most of that collective tail is redundancy that fewer, rescaled latents can carry; a circuit’s size is therefore meaningful only alongside whether its coefficients were fitted, and weighted and unweighted sizes are compared only at matched faithfulness (Figure 3).

Ablation methods are part of the method. Miller et al. (2024) showed that faithfulness is not robust to the choice of ablation; our measurements push the point into training. The home-turf effect (Section 5.3), node sets trained under different ablation methods sharing fewer latents than node sets trained for different objectives, and the miscalibration of coefficients trained without zero ablation (Section 4.3) all say the same thing: the ablation methods used for discovery, training, and evaluation belong in the method statement. Further discussion (equivalence classes, position handling, composing circuits into chains) is in Appendix J.

6.1 Limitations

The SAE basis is not ground truth. Many latents resist clean characterisation, and circuits built from ambiguous latents inherit that ambiguity; our scores measure intervention effects, not interpretability.

The objective is a surrogate. A high pre-activation does not guarantee a Top-K slot, so WCM trains on a continuous surrogate of a discrete event; the rank-keep term narrows the gap on training contexts but does not guarantee the target’s rank on held-out ones, and evaluation closes the loop by scoring the activation itself. For targets that barely enter the Top-K in the unmodified model the gap cannot be closed, which is why they are reported separately.

Faithfulness does not imply specificity. A circuit can restore its target by lifting the target’s whole neighbourhood; the rank-keep term limits the lifts that overtake the target, but does not reliably remove such circuits, so specificity is measured and reported rather than claimed.

Wiring is descriptive and depends on the ablation method. A weighted circuit is fitted as a set of upstream latents sufficient for one target; direct-effect edges wire it (Appendix N.4), but the wiring is dense, much of it cancels, and edge sets agree only partly across ablation methods, so only edges shared by all three are interpreted, and composing circuits into chains still needs its own validation.

Contrast contexts. Contrast contexts are verified silent only in the sense that the target stays out of its Top-K; a near-miss pre-activation can remain, and closeness is measured in one pooled representation space, so “close” is one of several defensible constructions.

Generality. TuringLLM’s synthetic corpus is an asset for control but a limit on generality, which the two public models partly address; larger models and other dictionary architectures remain open. The method is scoped to Top-K dictionaries by construction: on dense ReLU dictionaries, a random-circuit baseline with fitted coefficients loses its discriminating power once a set is large enough to span the anchors, so that architecture needs its own baseline design.

7 Conclusion

We reframed circuit discovery around a target latent, asking not which latents cause an output but what makes a latent fire. Weighted circuit masking answers with a weighted circuit of 10^2 – 10^3 upstream latents per target, tested on held-out contexts against random circuits given the same fitting; 46% of 15,046 sampled TuringLLM latents pass one pre-stated criterion [*pending: Pythia-70m and GPT-2 small, DAN-40/41*]; the coefficients locate a redundancy that costs $\sim 10^4$ latents at natural scale for deep targets, about $14\times$ the weighted circuit. Circuits for single latents are a building block: composing them into chains, with propagation tested at every link, is the path from dependencies to a map of the model’s internal computation.

Reproducibility

The full system (pipeline stages, evidence stores, the discovery methods of Section 4, the target-intervention evaluation, and the analysis code that produces every figure and table in this paper) is available at github.com/DanielJamesDavies/turing-explorer-circuit-discovery. TuringLLM, its synthetic training corpus, and the trained SAE bank are described in Appendix A; every experiment runs from a released configuration under the protocol of Appendix B, and each analysis reads only the run artifacts written by the pipeline stages.

References

- Marah Abdin, Sam Ade Jacobs, Ammar Ahmad Awan, Jyoti Aneja, Ahmed Awadallah, Hany Awadalla, Nguyen Bach, Amit Bahree, Arash Bakhtiari, Harkirat Behl, et al. Phi-3 technical report: A highly capable language model locally on your phone. *arXiv preprint arXiv:2404.14219*, 2024.
- Emmanuel Ameisen, Jack Lindsey, Adam Pearce, Wes Gurnee, Nicholas L. Turner, Brian Chen, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermyn, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam Zimmerman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. Circuit tracing: Revealing computational graphs in language models. *Transformer Circuits Thread*, 2025. <https://transformer-circuits.pub/2025/attribution-graphs/methods.html>.
- Jimmy Ba and Rich Caruana. Do deep nets really need to be deep? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2014.

- Adithya Bhaskar, Alexander Wettig, Dan Friedman, and Danqi Chen. Finding transformer circuits with edge pruning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E. Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.
- Diego Caples, Jatin Nainani, Callum McDougall, and rrenaud. Scaling sparse feature circuit finding to Gemma 9B. LessWrong, <https://www.lesswrong.com/posts/PkeB4TLxgaNnSmddg>, 2025.
- Arthur Conmy, Augustine N. Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. In *International Conference on Learning Representations (ICLR)*, 2024.
- Ruixuan Deng, Xiaoyang Hu, Miles Gilberti, Shane Storks, Aman Taxali, Mike Angstadt, Chandra Sripada, and Joyce Chai. Sparse feature coactivation reveals causal semantic modules in large language models. In *Proceedings of the Association for Computational Linguistics (ACL)*, 2026.
- Hendrik Droste, Christian Medeiros Adriano, Kathrin Korte, and Holger Giese. Where decoder cosine similarity fails for SAE feature flow discovery. *arXiv preprint arXiv:2609.12591*, 2026.
- Jacob Dunefsky, Philippe Chlenski, and Neel Nanda. Transcoders find interpretable LLM feature circuits. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Tom Dupre la Tour and Dan Mossing. Debugging misaligned completions with sparse-autoencoder latent attribution. <https://alignment.openai.com/sae-latent-attribution>, 2025.
- Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. Toy models of superposition. *Transformer Circuits Thread*, 2022. https://transformer-circuits.pub/2022/toy_model/index.html.
- Lucy Farnik, Tim Lawson, Conor Houghton, and Laurence Aitchison. Jacobian sparse autoencoders: Sparsify computations, not just activations. *arXiv preprint arXiv:2502.18147*, 2025.
- Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. *arXiv preprint arXiv:2406.04093*, 2024.
- Leo Gao, Achyuta Rajaram, Jacob Coxon, Soham V. Govande, Bowen Baker, and Dan Mossing. Weight-sparse transformers have interpretable circuits. *arXiv preprint arXiv:2511.13653*, 2025.
- Xuyang Ge, Fukang Zhu, Wentao Shu, Junxuan Wang, Zhengfu He, and Xipeng Qiu. Automatically identifying local and global circuits with linear computation graphs. *arXiv preprint arXiv:2405.13868*, 2024.
- Elena Golimblevskaia, Aakriti Jain, Bruno Puri, Ammar Ibrahim, Wojciech Samek, and Sebastian Lapuschkin. Circuit insights: Towards interpretability beyond activations. *arXiv preprint arXiv:2510.14936*, 2025.

- Wes Gurnee. SAE reconstruction errors are (empirically) pathological, 2024. AI Alignment Forum. <https://www.alignmentforum.org/posts/rZPiuFxESMxCdHe4B/sae-reconstruction-errors-are-empirically-pathological>.
- Tal Haklay, Hadas Orgad, David Bau, Aaron Mueller, and Yonatan Belinkov. Position-aware automatic circuit discovery. *arXiv preprint arXiv:2502.04577*, 2025.
- Michael Hanna, Sandro Pezzelle, and Yonatan Belinkov. Have faith in faithfulness: Going beyond circuit overlap when finding model mechanisms. In *Conference on Language Modeling (COLM)*, 2024.
- Zhengfu He, Xuyang Ge, Qiong Tang, Tianxiang Sun, Qinyuan Cheng, and Xipeng Qiu. Dictionary learning improves patch-free circuit discovery in mechanistic interpretability: A case study on Othello-GPT. *arXiv preprint arXiv:2402.12201*, 2024.
- Stefan Heimersheim and Neel Nanda. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, 2024.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- Andrej Karpathy. Let’s reproduce GPT-2 (124m), 2024. <https://github.com/karpathy/build-nanogpt>.
- Dmitrii Kharlapenko, Stepan Shabalin, Fazl Barez, Arthur Conmy, and Neel Nanda. Scaling sparse feature circuit finding for in-context learning. *arXiv preprint arXiv:2504.13756*, 2025.
- János Kramár, Tom Lieberum, Rohin Shah, and Neel Nanda. AtP*: An efficient and scalable method for localizing LLM behaviour to components. *arXiv preprint arXiv:2403.00745*, 2024.
- Daniil Laptev, Nikita Balagansky, Yaroslav Aksenov, and Daniil Gavrilov. Analyze feature flow to enhance interpretation and steering in language models. *arXiv preprint arXiv:2502.03032*, 2025.
- Patrick Leask, Bart Bussmann, Michael Pearce, Joseph Bloom, Curt Tigges, Noura Al Moubayed, Lee Sharkey, and Neel Nanda. Sparse autoencoders do not find canonical units of analysis. In *International Conference on Learning Representations (ICLR)*, 2025.
- Michael A. Lepori, Thomas Serre, and Ellie Pavlick. Uncovering intermediate variables in transformers using circuit probing. *arXiv preprint arXiv:2311.04354*, 2023.
- Maximilian Li and Lucas Janson. Optimal ablation for interpretability. *arXiv preprint arXiv:2409.09951*, 2024.
- Yuxiao Li, Eric J. Michaud, David D. Baek, Joshua Engels, Xiaoqing Sun, and Max Tegmark. The geometry of concepts: Sparse autoencoder feature structure. *Entropy*, 27(4):344, 2025.
- Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L. Turner, Craig Citro, et al. On the biology of a large language model. *Transformer Circuits Thread*, 2025. <https://transformer-circuits.pub/2025/attribution-graphs/biology.html>.
- Christos Louizos, Max Welling, and Diederik P. Kingma. Learning sparse neural networks through L_0 regularization. In *International Conference on Learning Representations (ICLR)*, 2018.
- Alireza Bayat Makou, Jingcheng Niu, Subhabrata Dutta, and Iryna Gurevych. Many circuits, one mechanism: Input variation and evaluation granularity in circuit discovery. *Transactions on Machine Learning Research*, 2026.
- Samuel Marks, Can Rager, Eric J. Michaud, Yonatan Belinkov, David Bau, and Aaron Mueller. Sparse feature circuits: Discovering and editing interpretable causal graphs in language models. In *International Conference on Learning Representations (ICLR)*, 2025.

- Joseph Miller, Bilal Chughtai, and William Saunders. Transformer circuit faithfulness metrics are not robust. In *Conference on Language Modeling (COLM)*, 2024.
- Aaron Mueller. Missed causes and ambiguous effects: Counterfactuals pose challenges for interpreting neural networks. *arXiv preprint arXiv:2407.04690*, 2024.
- Aaron Mueller, Atticus Geiger, Sarah Wiegrefe, Dana Arad, Iván Arcuschin, Adam Belfki, Yik Siu Chan, Jaden Fiotto-Kaufman, Tal Haklay, Michael Hanna, Jing Huang, Rohan Gupta, Yaniv Nikankin, Hadas Orgad, Nikhil Prakash, Anja Reusch, Aruna Sankaranarayanan, Shun Shao, Alessandro Stolfo, Martin Tutek, Amir Zur, David Bau, and Yonatan Belinkov. MIB: A mechanistic interpretability benchmark. In *International Conference on Machine Learning (ICML)*, 2025.
- Neel Nanda. Attribution patching: Activation patching at industrial scale. <https://www.neelnanda.io/mechanistic-interpretability/attribution-patching>, 2023.
- Chris Olah, Nick Cammarata, Ludwig Schubert, Gabriel Goh, Michael Petrov, and Shan Carter. Zoom in: An introduction to circuits. *Distill*, 2020. <https://distill.pub/2020/circuits/zoom-in>.
- Charles O’Neill and Thang Bui. Sparse autoencoders enable scalable and reliable circuit identification in language models. *arXiv preprint arXiv:2405.12522*, 2024.
- Vedant Palit, Florent Draye, Terry Jingchen Zhang, Bernhard Schölkopf, and Zhijing Jin. How do linear probes emerge? a circuit-tracing framework with concept-targeted attribution. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.
- Gonçalo Paulo and Nora Belrose. Evaluating SAE interpretability without explanations. *arXiv preprint arXiv:2507.08473*, 2025.
- Gonçalo Paulo, Alex Mallen, Caden Juang, and Nora Belrose. Automatically interpreting millions of features in large language models. *arXiv preprint arXiv:2410.13928*, 2024.
- Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners, 2019. OpenAI Technical Report.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.
- Prajit Ramachandran, Barret Zoph, and Quoc V. Le. Searching for activation functions. *arXiv preprint arXiv:1710.05941*, 2017.
- Pedro Savarese, Hugo Silva, and Michael Maire. Winning the lottery with continuous sparsification. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Noam Shazeer. GLU variants improve transformer. *arXiv preprint arXiv:2002.05202*, 2020.
- Claudia Shi, Nicolas Beltran-Velez, Achille Nazaret, Carolina Zheng, Adrià Garriga-Alonso, Andrew Jesson, Maggie Makar, and David M. Blei. Hypothesis testing the circuit hypothesis in LLMs. *arXiv preprint arXiv:2410.13032*, 2024.
- Lewis Smith, Senthooan Rajamanoharan, Arthur Conmy, Callum McDougall, János Kramár, Tom Lieberum, Rohin Shah, and Neel Nanda. Negative results for sparse autoencoders on downstream tasks and deprioritising SAE research, 2025. Google DeepMind Mechanistic Interpretability Team progress update. <https://deepmindsafetyresearch.medium.com/negative-results-for-sparse-autoencoders-on-downstream-tasks-and-deprioritising-sae-research-6cadcf125b9>.
- Niklas Stoehr, Kevin Du, Vésteinn Snæbjarnarson, Robert West, Ryan Cotterell, and Aaron Schein. Activation scaling for steering and interpreting language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, 2024.

- Aaquib Syed, Can Rager, and Arthur Conmy. Attribution patching outperforms automated circuit discovery. *arXiv preprint arXiv:2310.10348*, 2023.
- Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L. Turner, Callum McDougall, Monte MacDiarmid, C. Daniel Freeman, Theodore R. Sumers, Edward Rees, Joshua Batson, Adam Jermy, Shan Carter, Chris Olah, and Tom Henighan. Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet. *Transformer Circuits Thread*, 2024. <https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html>.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. LLaMA: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.
- Vladimir Vorobev and Maxim Kuznetsov. A paraphrasing model based on ChatGPT paraphrases, 2023. https://huggingface.co/humarin/chatgpt_paraphraser_on_T5_base.
- Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *International Conference on Learning Representations (ICLR)*, 2023.
- Naiyu Yin, Dennis Wei, Tian Gao, Amit Dhurandhar, Karthikeyan Natesan Ramamurthy, and Yue Yu. Scalable circuit learning for interpreting large language models. *arXiv preprint arXiv:2606.16939*, 2026.
- Biao Zhang and Rico Sennrich. Root mean square layer normalization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

A Corpus and Model Details

A.1 A Corpus Built by Design

The synthetic corpus was constructed in two stages. The first produced a relatively small but high-quality foundation: Phi-3 Mini (Abdin et al., 2024), a 3.8B-parameter language model available under a permissive licence, generated instructional and educational text spanning 18 broad subject areas (Physics, Mathematics, Computer Science, Chemistry, Biology, Philosophy, English, Creative Writing, Languages, Art, Music, Engineering, Economics, Business, Geography, History, Culinary Arts, and Communication), recursively generating topics and sub-topics and producing structured documents of headings and text.

The second stage addressed scale. A fine-tuned T5-based paraphrase model (Vorobev and Kuznetsov, 2023; Raffel et al., 2020) expanded the corpus by generating 11 paraphrased variants of each source document, motivated by knowledge distillation results (Ba and Caruana, 2014; Hinton et al., 2015) suggesting that varied outputs of a stronger teacher can be advantageous over raw data alone. The final corpus totals approximately 2 billion tokens.

Two tokenized datasets were derived using the Phi-3 Mini tokenizer. The language-modelling training set combines original and augmented texts shuffled into epochs, concluding with a second pass over only the high-quality originals to consolidate representations around the cleaner signal. The interpretability dataset consists of 64-token chunks shuffled into 4,096 shards, allowing the pipeline to process one shard at a time with a bounded memory footprint.

A.2 Architecture and Training

TuringLLM follows the residual-stream design of the GPT family (Radford et al., 2019) with two deliberate departures: SwiGLU (Shazeer, 2020; Ramachandran et al., 2017) replaces the ReLU MLP,

Attribute	Value
Parameters	254M
Layers	12
Residual stream dimension	1,024
Attention heads	16
MLP hidden dimension	4,096
Context length	1,024 tokens
Vocabulary size	50,304 tokens
MLP activation	SwiGLU
Normalisation	RMSNorm
Tokenizer	Phi-3 Mini

Table 3: TuringLLM-1.0-254M architecture summary.

following the motivation of the Llama family (Touvron et al., 2023), and RMSNorm (Zhang and Sennrich, 2019) replaces layer normalisation. Table 3 lists its dimensions and Figure 4 sketches the architecture. Training reached a minimum cross-entropy of approximately 1.62 (train) and 1.80 (validation); the same infrastructure applied to GPT-2 (124M) reaches approximately 3.29 (Karpathy, 2024), reflecting the tighter distribution of the synthetic corpus. The loss curve (Figure 5) drops sharply in its final phase, corresponding to the second pass over Phi-3 Mini originals.

A.3 Fitness for Interpretability

TuringLLM is not designed to win benchmarks; it is designed to be a controlled subject. Standard held-out benchmarks fit poorly for a base model trained on synthetic educational text, so fitness was evaluated qualitatively and through structured internal tests: Phi-3 Mini composed topic-specific opening sentences and scored TuringLLM’s completions on English quality and topic adherence, yielding average grades of approximately B and A respectively across the 18 subject areas. This level of competence supports the working hypothesis that the model contains structured internal representations worth studying; a model merely interpolating noise would not produce consistently topical completions. Representative completions are shown in Figure 7.

A.4 Sparse Autoencoder Quality

Figure 6 reports three standard quality metrics for the 36-SAE bank, per site. *Explained variance*, $1 - \text{Var}(x - \hat{x}) / \text{Var}(x)$, measures reconstruction fidelity. *Alive latents* is the fraction of each dictionary’s 40,960 latents that fired at least once during the full collection pass over the interpretability corpus. *Loss recovered* splices each SAE’s reconstruction into the model at its site and reports the recovered share of the cross-entropy gap between the original model and zero-ablating that site (Marks et al., 2025), measured over 12,288 tokens of the interpretability corpus. Because the SAEs are Top-K with $K = 128$, sparsity is fixed by construction and the usual L_0 –fidelity trade-off curve is a single point per SAE.

Three observations matter for interpreting the paper’s results. First, reconstruction fidelity is healthy across the bank: attention SAEs explain 0.93–0.99 of variance, residual SAEs decline smoothly with depth to 0.85, and MLP SAEs bottom out at 0.77 in layer 9. Second, the layer-1 attention SAE has a dead-latent crater: only 37% of its latents ever fire (84% at layer 2, at least 93% everywhere else). Third, loss recovered exposes a functional gap that explained variance misses: the layer-11 attention SAE recovers only 55% of the cross-entropy gap despite explaining 0.96 of variance (layer 10: 84%), indicating that the unexplained residual at the final attention site is disproportionately important to the logits.

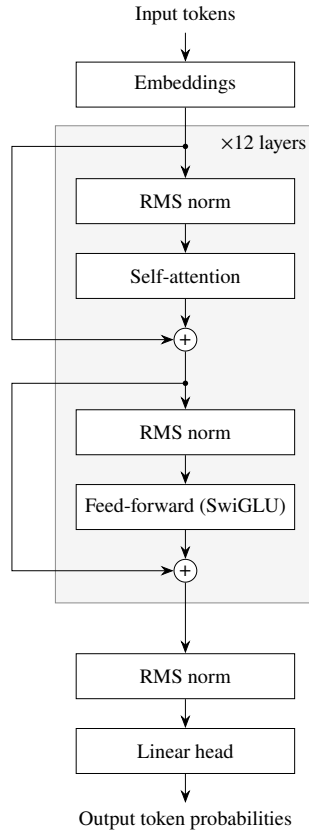


Figure 4: TuringLLM-1.0-254M architecture: a 12-layer pre-norm decoder-only transformer with RMSNorm and a SwiGLU feed-forward block. The circuit-discovery pipeline attaches SAEs at three sites per layer: attention output, MLP output, and the residual stream.



Figure 5: TuringLLM training and validation cross-entropy over the 3,985-step run (log scale). The dashed reference marks the validation loss reached by the same training infrastructure applied to GPT-2 (124M); the drop near step 3,700 corresponds to the final pass over only the high-quality Phi-3 Mini originals.

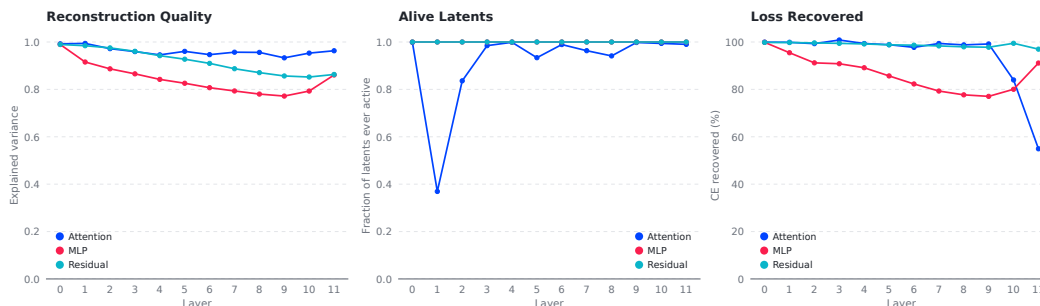


Figure 6: Quality of the 36-SAE bank by site. Left: explained variance of each SAE’s reconstruction. Middle: fraction of latents that fired at least once over the full collection pass. Right: percentage of the cross-entropy gap recovered when splicing each SAE’s reconstruction into the model, relative to zero-ablating the site.

“Physics utilizes the fundamental concepts of quantum mechanics to explain how electrons populate a sublevel within an atom, which is marked by energy levels. These levels define the overall potential energy and enable electrons to occupy specific energy levels or sublevels [. . .]”

“Artificial Intelligence (AI) has played a pivotal role in elevating personalization in human-robot interfaces, elevating the potential for truly personalized and responsive user experiences. The evolution towards AI extends its roots back to the 1960s with early rule-based [. . .]”

“Artificial Intelligence (AI) and Machine Learning (ML) technology have become indispensable in modern engineering. These technologies enable engineers to analyze vast datasets with unprecedented efficiency, leading to significant improvements in design processes, maintenance, and optimization.”

Figure 7: Representative TuringLLM completions (excerpts). Continuations are topical and grammatical, supporting the use of the model as an interpretability subject; occasional repetitions and factual slips (e.g. conflated terminology) reflect its 254M-parameter scale and synthetic corpus.

B Protocol

Every circuit in this paper is discovered and scored under the settings in Table 4, on every model; only the model and its dictionary change.

C Attribution Baselines and Position-Aware Node Sets

C.1 Attribution Objectives

Two gradient objectives supply the signed scores, one for each side of the target’s behaviour.

Contrastive gradient: why does the target not fire? When a latent is absent from the Top-K code its activation is identically zero and a gradient through it carries no information, so attribution targets the target latent’s *encoder pre-activation*

$$\hat{a}_f(x, t) = \langle W_e^{(f)}, z(x, t) \rangle + b_e^{(f)},$$

with $W_e^{(f)}$ the target’s encoder row, $b_e^{(f)}$ its effective bias, and $z(x, t)$ the pre-SAE dense activation. The pre-activation is defined on every forward pass whether or not f clears Top-K, and is differentiable with respect to all upstream latents through identity passthroughs at each SAE site. On a contrast context x , the objective is the squared gap $\mathcal{L}^{\text{ctr}}(x) = (\hat{a}_f(x, t^*) - \mu_f^+)^2$ to the target’s mean pre-activation over activating contexts, taken at the position of maximum pre-activation, and each upstream latent g is scored by gradient-times-activation averaged over contrast contexts: $S_g^{\text{ctr}} = \frac{1}{|N_f|} \sum_x a_g(x, t_g) \partial \mathcal{L}^{\text{ctr}}(x) / \partial a_g(x, t_g)$. The baseline keeps the top-scoring latents of each sign of S_g^{ctr} (Algorithm 1). The source of contrast contexts is a controlled variable: `close` (the retrieved store), `random`, or `distant`.

Ablation gradient: what sustains the firing? The mirror score on activating contexts, $S_g^{\text{ab}} = \frac{1}{|\mathcal{P}_f|} \sum_{(x, t)} a_g(x, t_g) \partial a_f(x, t) / \partial a_g(x, t_g)$, is large when the upstream latent is strongly active and the target is sensitive to it; its top latents form the node set.

Hybrid fusion. A hybrid arm runs both objectives on the same target and fuses the circuits by latent identity, each node keeping its per-source scores; fusion details, pruning, and acceptance modes are in Appendix K and Algorithm 3.

Co-activation graph. A second pass over stored top contexts computes, for every latent, its top 64 co-activating partners scored by clamped pointwise mutual information. In this paper it serves as the correlational reference against which the causal discoveries are compared (Appendix F).

C.2 Attribution Modes

Both objectives are estimator-agnostic: evaluating the gradient at the clean activations is one choice of *linearisation point*, not the only one. Four modes differ only in where gradients are linearised, holding objective, probes, and selection fixed:

- **local**: a single linearisation at the natural activations (gradient-times-activation at the clean forward pass), descending from attribution patching (Nanda, 2023; Syed et al., 2023).
- **ig_mean**: integrated gradients along the straight path from the site’s mean-ablation value to the natural activations; this is the IE_{ig} node estimator of Marks et al. (2025), which we adopt unchanged and credit as theirs.
- **restoration**: iterated greedy re-linearisation, in which each round attributes at the *current* partially restored stream, admits the top scorers, restores them by live re-encoding, and re-linearises, for as many rounds as the target has upstream sites (Algorithm 4).

- **ig_restoration**: the restoration loop with per-round integrated-gradient scoring.

The taxonomy is single-point, path, and iterated-trajectory linearisation. Adopting the published path estimator is deliberate experimental design: comparisons between output- and latent-targeted discovery (Section 5.4) then isolate the target and the node-set rule rather than estimator quality, exactly the claim under test.

C.3 Node Sets and Position-Aware Discovery

Attribution returns ranked latents, so the baselines must also decide which node set a ranking defines. The Top-K constraint makes the classical answer fail structurally: the code is k -sparse *per position* while a target reads its whole causal prefix, so a position-collapsed circuit must cover the union of every position’s active set, thousands of distinct latents per site, or recomputed faithfulness collapses. The failure is a property of position aggregation, not of node selection (mechanism measured in Appendix M.1).

The baselines’ node set is therefore a set of latents, the same kind of object (P) optimises over, collected across positions. Latents outside the circuit are ablated everywhere; nodes are not forced to any value: they are recomputed from the modified stream at every position, so positions instantiate naturally (allow, don’t force), exactly the circuit-only execution of Section 4.1. Discovery is position-aware to match: attribution is computed per position, a per-target percentile cut on $|\text{attribution}|$ is applied per position, and the node set is the union over the positions of the target’s causal prefix. No template or cross-example alignment is required, and in our sweeps the percentile rule dominates top- N and per-position-normalised alternatives. Both attribution signs are kept, as a design principle rather than an implementation detail: the node set is *stream reconstruction*, not a selection of what drives the target, and a negative-attribution latent still carries part of the residual stream that nodes are recomputed from, so removing it corrupts recomputed execution.

Two optional, validated levers shrink the set: a *recurrence prune* (nodes must fire in at least two contexts) and a *magnitude-bisection prune* (a binary search over the attribution ranking for the smallest prefix holding a requested faithfulness, automating the threshold-tuning step of prior practice Marks et al., 2025); details and their depth-dependence are in Appendix N.

A node set’s size at a stated faithfulness is a measured quantity, and Appendix D argues it measures the structure of the computation rather than a defect of the method.

D Control versus Faithfulness at Natural Scale

Which latents can impose a target’s activation? A few hundred. How many must remain for the circuit alone to reproduce it? Tens of thousands. The two questions are two criteria over the same node set. *Interventional control* asks for a small set whose intervention scores clear an acceptance gate, sufficiency to induce $\phi_{\text{ind}} \geq \tau_{\text{ind}}$ and necessity $\phi_{\text{sup}} \geq \tau_{\text{sup}}$, as the gradient attribution methods of Appendix K used. *Faithfulness* is (P) with every coefficient fixed at $\alpha = 1$: the smallest node set that reproduces the target alone, whose size n_ϵ we call the target’s *faithfulness cost*. The clamped/recomputed decomposition separates them, and on depth-spanning target panels their answers come apart; this appendix measures how far, for node sets from the attribution baselines of Appendix C. All numbers here come from panels of three to four targets per layer under the evaluation that preceded protocol v1, and are indicative.

Control is concentrated. Restricting a node set to its most influential positions barely moves clamped faithfulness: it reaches 0.79–1.04 with only 10% of positions kept, roughly flat as more are added. What *causally controls* a target lives in a handful of positions and a compact set of latents: a circuit in the classic, hand-annotatable sense. The running example’s controlling circuit reads like a chemistry syllabus: latents for Arrhenius kinetics, the third law, and absolute zero.

Faithfulness is collective, position-local, and depth-scaled. Recomputed faithfulness tells the opposite story. At the same 10% of positions a shallow target still passes (0.85), but deep targets



Figure 8: Control is sparse; faithfulness is collective; scaling coefficients collapse its cost. Cumulative score against nodes added in attribution order (three targets per depth band: median curves, individual targets ghosted; position-aware path attribution, percentile selection). Clamped faithfulness (node selection) rises steeply within 10^2 – 10^3 nodes, while recomputed faithfulness (the circuit alone, at $\alpha = 1$) sits near zero until 10^4 – 10^5 nodes, under mean ablation (solid) and zero ablation (dashed). The shaded clamped–recomputed gap widens with target depth. Diamonds: *weighted circuits* discovered on the same targets (Section 5.2), scored with their scaling coefficients applied under zero ablation, a few hundred nodes reaching the score the $\alpha = 1$ curve needs 10^5 for.

sit at exactly 0.00 and climb only position by position as the rest of the prefix is restored (a layer-7 residual-stream target: $0 \rightarrow 0.24 \rightarrow 0.52 \rightarrow 0.82$). This is the sense in which the cost is *position-local*: no small set of positions carries it, and the score accrues position by position across the causal prefix. The faithfulness-versus-size curve (Figure 8) is two-regime on every target we measured: most compression is available down to $\phi^{\text{free}} \approx 0.7$, then a hard limit, an irreducible core that even $\phi^{\text{free}} = 0.2$ requires. That core is the faithfulness cost n_ε at loose tolerance, and it scales with depth: roughly 2k at layer 3, 10k at layer 7, 60k at layer 11. Compressibility follows the same gradient: shallow node sets prune to 13–23% of their size, deep ones only to $\sim 70\%$ with a sharp cliff. Depth, not the selection rule, is the fundamental determinant of circuit size. These are costs *at natural scale* ($\alpha = 1$): Section 5.2 shows that fitted per-node scaling coefficients collapse them by a factor of ten to a hundred.

Individually negligible, collectively load-bearing. Leave-one-out is blind to all of this: removing any single node changes the target by 0.000, the noise level. Yet cutting nodes in bulk by their own attribution ranking craters function: on recurrence- and bisection-validated node sets, dropping the bottom half by score moves ϕ_Z^{free} from 0.975 to 0.895 at layer 2 but from 1.119 to 0.595 at layer 8; dropping the bottom 90% yields 0.635, 0.000, 0.222, and 0.005 at layers 2, 8, 9, and 10. The scores being cut are minuscule, 2 – 5×10^{-4} in attribution units. Nodes individually at the noise level collectively carry the entire deep reconstruction. Individual-contribution thresholding, the standard selection rule across circuit discovery, therefore systematically undercounts what faithfulness needs; this is the finding that makes circuit size a measurement rather than a defect.

The union survives its controls. Node sets are target-specific, not noise or soup (cross-target node-set tails overlap at chance, Jaccard ~ 0.07 , against 0.91 for a target’s own split halves); the plain magnitude union Pareto-dominates every structured alternative we tried; and at depth, faithfulness is partly carried by the preserved SAE errors, the latent-targeted echo of the error-node finding for output targets (Marks et al., 2025). The full controls are in Appendix N.

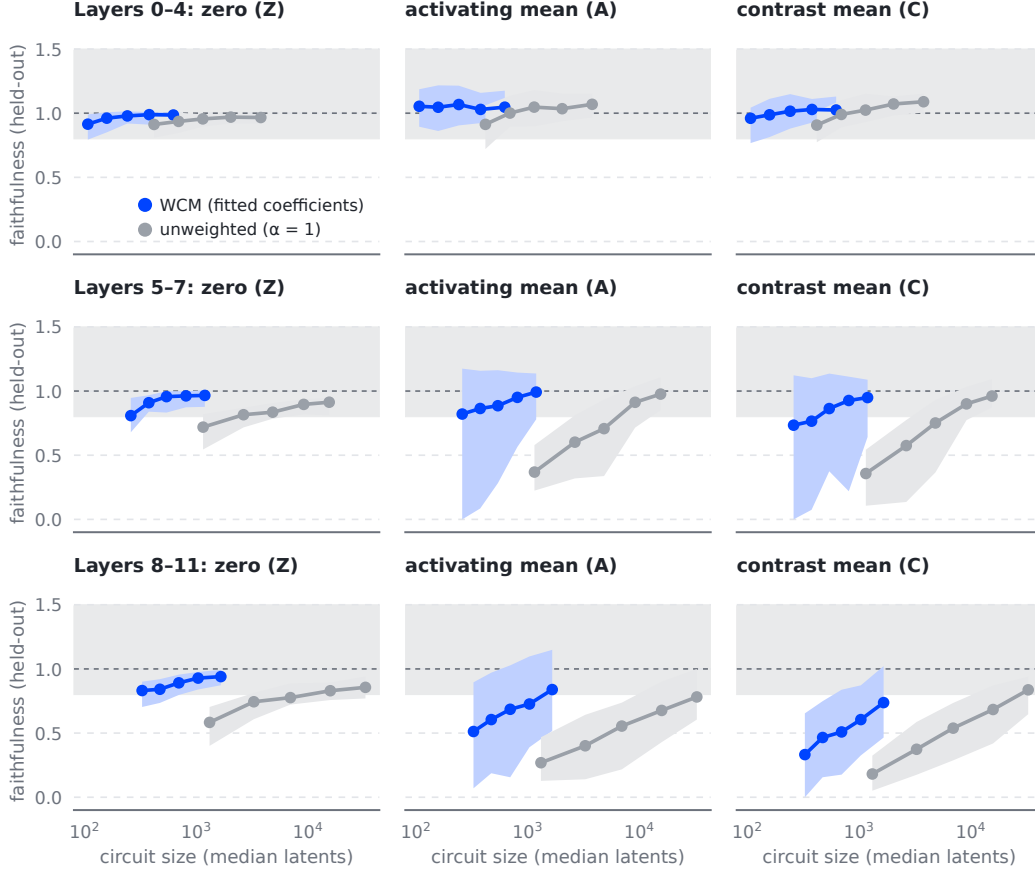


Figure 9: Weighted against unweighted circuit masking by ablation method (rows: target depth band; columns: zero ablation, and mean ablation over activating and over contrast contexts). Median held-out faithfulness and interquartile range against median circuit size, one point per sparsity price; the grey band is $[0.8, 1.25]$. The gap between the methods is smallest under zero ablation, largest under the two mean ablations, and widens with depth.

E Weighted Circuits: Sufficiency to Induce and the Training Budget

Weighted against unweighted circuits across the sparsity price. The body comparison (Figure 3) uses the one price both grids share, $\lambda = 10^{-3}$; the natural-scale cost comes from the full sweep, which fits WCM at $\lambda \in \{2.5, 5, 10, 20, 40\} \times 10^{-4}$ and unweighted circuit masking at $\lambda \in \{0.1, 0.3, 1, 3, 10\} \times 10^{-4}$ to 191 targets (five per layer and site kind, plus the pilot targets; 190 with usable contexts) and scores every circuit on the held-out strongest contexts. Unweighted masking needs a lower price to reach comparable faithfulness, so the two grids differ. Figure 9 separates the three ablation methods: the gap is smallest under zero ablation and largest under the two mean ablations, where unweighted circuits of $\sim 10^3$ latents score 0.18–0.37 at layers 5–11 (zero ablation 0.58–0.72). For the natural-scale cost we take, per target, the smallest circuit whose worst score lies within 0.3 of 1. Among targets both methods reach, the medians are 128 against 502 latents at layers 0–4 (3.7 $\times$; 66 of 76 targets reached by both), 447 against 3,862 at 5–7 (9.8 $\times$; 33 of 48), and 789 against 9,989 at 8–11 (14.1 $\times$; 25 of 66). WCM’s grid stops near 10^3 latents, so at depth it reaches the criterion for fewer targets (31 of 66) than unweighted masking does at up to 3×10^4 latents (39 of 66); *[pending: WCM at λ below 2.5×10^{-4}].*

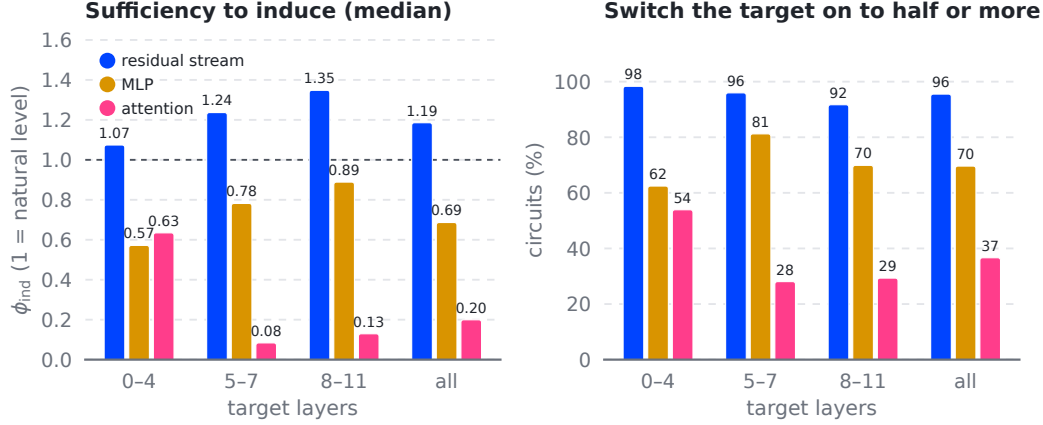


Figure 10: Sufficiency to induce across the production run (15,004 TuringLLM circuits), on held-out contrast contexts where the target is silent. Left: median $\phi_{\text{ind}}^{\alpha}$ by target depth band and site kind (1 is the target’s natural level, dashed). Right: the share of circuits that switch the target on to at least half its natural level. Reported for every circuit; not part of the pass criterion.

Sufficiency to induce by site kind. Sufficiency to induce, unlike reconstruction, is structured by the target’s site kind (Figure 10). It asks a question no circuit is trained on: setting the circuit’s latents to α times their values from the activating contexts, does the target switch on in held-out contrast contexts where it is silent? Across the production run the median is 0.84 (0.98 among passing circuits, of which 88% switch the target on to at least half its natural level). Residual-stream targets are induced almost everywhere (96% at half or more; median 1.19, rising with depth to 1.35 at layers 8–11), MLP-output targets partly (70%; median 0.69), and attention-output targets rarely beyond layer 4 (28% and 29% at layers 5–7 and 8–11; median 0.20 overall). Which latents *can* be induced from upstream is therefore itself a per-kind property, and characterising it is open.

The budget is part of the method. Sweeping the training budget shows reconstruction faithfulness converges by 100 steps; everything after buys sparsity. That sparsity is paid for in the calibration of sufficiency to induce: on the panel, deep targets are induced with overshoot at the full budget (median $\phi_{\text{ind}}^{\alpha}$ 1.48) and calibrate at a quarter of it (1.10 deep, 1.01 shallow), at two to three times the circuit size. The sharpest case replicates held-out. A layer-9 circuit that reconstructs perfectly yet barely induces its target at the full budget (0.17) *induces* it (0.83) when discovered at a quarter of it; the calibration sweep shows the nodes that carry the induction being progressively pruned as training extends (0.88 $\rightarrow$ 0.57 $\rightarrow$ 0.25). Sparsity bought by budget and sparsity bought by price are also *different circuits*: at matched size the two prunings share only about half their nodes, a fifth non-identifiability probe (Section 5.5). Two further results on the training ablation methods close the loop. Coefficient-only training on a frozen node set does not fix the induction overshoot but does inflate the coefficients monotonically; the objective simply does not constrain sufficiency to induce. Removing the zero-ablation term from training explodes zero-ablation faithfulness to 36–1,681 at depth while the trained ablation method stays in band: the sharpest demonstration of the *home-turf effect* (Section 5.3), now at the level of coefficients rather than node sets. The training schedule therefore belongs in the method statement: the protocol fixes the full budget, and a quarter of it (at a raised price) calibrates sufficiency to induce better at two to three times the circuit size.

Training dynamics at scale. Every production fit records its per-step curve (Figures 11 and 12). All gates start open, so no latent falls below the keep threshold for the first ~ 55 steps; membership then collapses from 10^5 – 10^6 candidates to a median of about 1,700 by step 100, and a circuit is within twice its final size by a median step of 137. It is not converged at the protocol’s 400 steps: the median

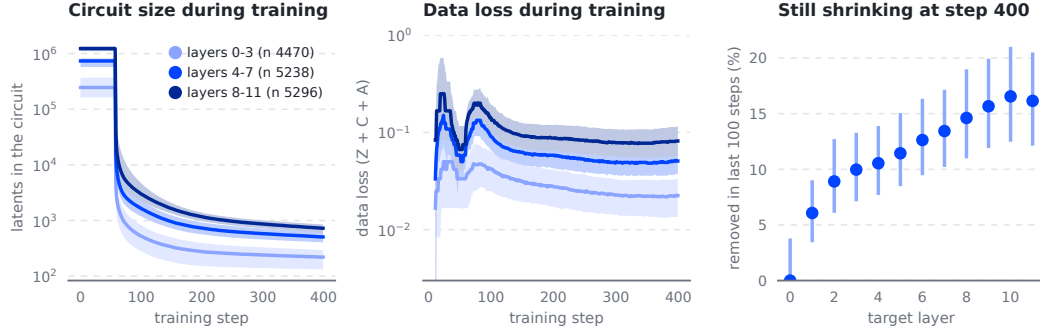


Figure 11: Training dynamics of the 15,004 production fits, by target depth band (median and interquartile range). Left: circuit size (latents above the keep threshold) per step. Middle: the data loss (the Z, C, and A terms, penalty excluded), averaged over each pass through the 48 training contexts (12 steps), since every step sees one batch of 4. Right: the share of members removed over the last 100 steps, by target layer (median and interquartile range).

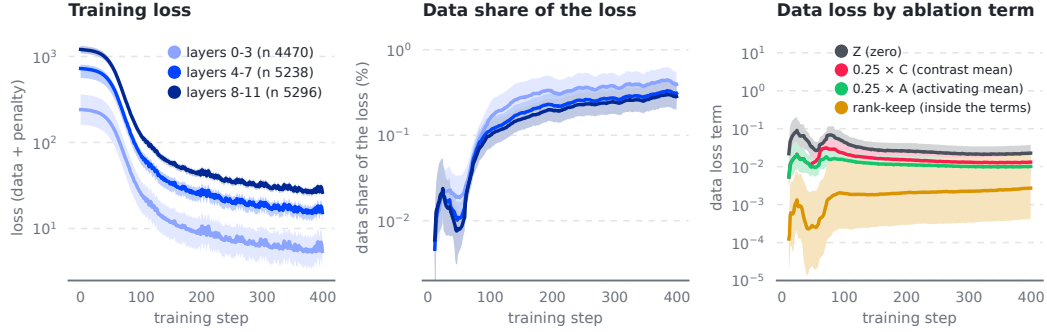


Figure 12: The training loss of the production fits (median and interquartile range; per-pass averages as in Figure 11). Left: the total loss, data terms plus sparsity penalty, by target depth band. Middle: the data terms' share of it; the rest is the penalty on open gates. Right: the data loss split into its ablation terms as weighted in the objective (Z, $0.25 \times C$, $0.25 \times A$), with the rank-keep contribution that each term contains.

circuit loses 12% of its members over the last 100 steps (16–17% at layers 10 and 11), and 88% are still shrinking by more than 5%. The data loss stays nearly flat over that stretch while the penalty falls, so the late pruning costs little training fit, but at depth the circuits still shrinking pass less often: 30% against 67% for the settled half at layers 4–7, and 9% against 36% at layers 8–11, with no difference at layers 0–3. Reported sizes are therefore sizes at a fixed budget. The sparsity penalty dominates the loss throughout: its tail of nearly closed gates outweighs the members, and the data terms end at 0.3% of the loss. Before their 0.25 weights, the contrast-mean and activating-mean errors are about twice the zero-ablation error, the same ordering as held-out faithfulness.

Attribution-mass recovery. Ge et al. (2024) score circuits by attribution recovery inside their linearised graph, with no intervention; our first-order analogue is *attribution-mass recovery*, the share of total activation-times-gradient mass a node set captures (a surrogate: the exact leaf-sum identity exists only in their substitute model). Under it the ranking of Table 2 inverts: plain top- n attribution leads (median signed recovery 0.76), the replication follows (0.72), and the weighted circuit trails (0.52) while being the only arm that passes the intervention test. The home-turf effect therefore

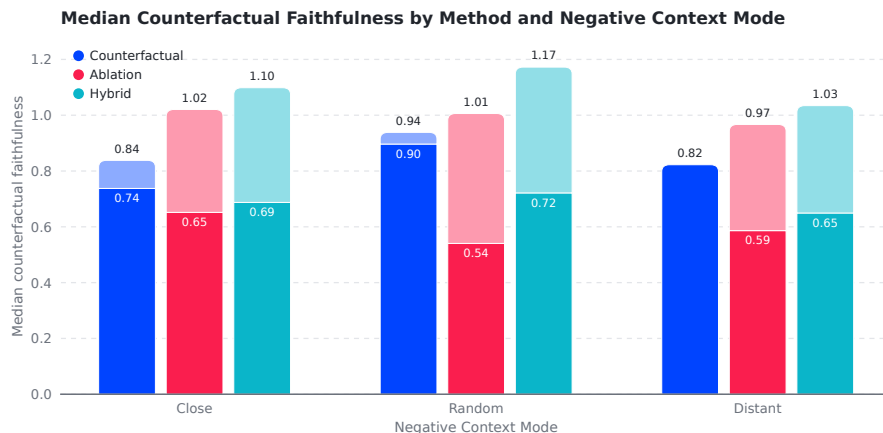


Figure 13: Median sufficiency to induce by discovery method and contrast-context source. Solid segments are medians over each method–source group’s accepted circuits; light segments extend to the median of the group’s 44 highest-scoring circuits, 44 being the smallest accepted count among the nine groups, for an equal-population comparison. Sources are *close* (the retrieval store of Appendix L, not checked for silence), *random* (uniformly sampled sequences), and *distant* (sequences maximally dissimilar to the activating contexts); this grid predates the silence-checked selection that WCM uses.

reaches evaluation frameworks themselves, and the gap carries a finding: a circuit that reproduces the target on held-out intervention while holding a third of the absolute attribution mass (0.37 median) shows that first-order attribution mass is not what faithfulness is made of.

F Dependencies Are Not Recoverable from Co-activation

If the discovered dependencies were merely co-occurrence in disguise, the co-activation graph would predict them. It does not. In a full-scale run of the gradient methods producing 7,525 accepted circuits, the mean overlap between a circuit’s discovered nodes and its target’s top co-activation neighbourhood was approximately 3.6%, the mean pairwise co-activation density among discovered nodes was approximately 0.26%, and neither quantity correlated with sufficiency to induce (Pearson $r = 0.02$; distributions in Figure 14). Latent co-activation profiles across the bank are themselves nearly orthogonal (the first two principal components explain roughly 3% of variance), so local co-occurrence structure carries little of the information that gradient-based contrast discovers. Causally validated dependency neighbourhoods, in short, are not local co-activation neighbourhoods.

The converse test is to let co-activation structure propose circuits directly. Under the same 128-target protocol, evidence base, and target-intervention gates, two correlational baselines produced no accepted circuits at all, while ablation gradient accepted circuits for 121 of the same 128 targets (Table 6). Sparse co-activation expansion follows the strongest co-activation edges outward from the target; differential activation ranks latents by their activation contrast between activating and contrast contexts. Correlational proposals do not merely score lower; they fail the intervention gates outright. The strongest version of this challenger — anchor co-activation selection granted the weighted circuit’s own coefficient fitting — revives on reconstruction but fails the joint calibration suite on every target tested, and is treated in Appendix P.2.

G Method and Contrast-Context Source

On the attribution grid (Figure 13), contrastive gradient achieves the highest median sufficiency to induce across all three sources, peaking at approximately 0.90 with random contrast contexts. Its advantage over ablation gradient is largest in the random condition (0.36), intermediate in the distant

condition (0.24), and smallest in the close condition (approximately 0.09). This ordering makes mechanistic sense: contrastive gradient is explicitly optimised for sufficiency to induce, while ablation gradient targets necessity and achieves sufficiency to induce as a by-product.

The effect of the contrast-context source is itself a finding. One might expect close contrast contexts, semantically similar to the activating contexts but lacking the target’s activation, to produce the sharpest contrastive signal. In practice the reverse holds for contrastive gradient: random contrast contexts outperform close ones by approximately 0.16 median points of sufficiency to induce. The most likely explanation is that close contrast contexts, being near the activating distribution, partially activate the target’s mechanism, providing a noisy gradient signal (this grid used the retrieval store, whose contrast contexts are not checked for silence; Appendix L), while random contrast contexts, where the target is more completely absent, produce a cleaner gap and a more reliable attribution. Semantic closeness and clean causal contrast, in other words, are not the same property, and contrast-context construction should be treated as an experimental variable rather than a solved preprocessing step.

Contrast source for WCM. For weighted circuit masking the source matters less. We refitted the same targets with silence-checked close, random, and distant contrast contexts (Appendix L) and with the store, and scored every fit against every source. Node sets move no more between sources than between two refits with the same source, and the headline faithfulness scores move by a few hundredths; what the source changes is robustness off the training distribution, with each fit most faithful under the contrast-context mean it was trained with. Close contrast contexts are the protocol default (Table 4); *[pending: the train-source $\times$ evaluation-source matrix at 288 targets, DAN-64]*.

H Specificity

Faithfulness, necessity, and the random-circuit baseline all read the target alone. A circuit can therefore restore its target by lifting the target’s whole neighbourhood, its concept, topic, or context type, and pass all three: a concept amplifier rather than a mechanism for that latent. The loss permits it, since reproducing the target’s pre-activation constrains the site’s stream along one direction of d_{model} , and it leaks into the activation read only when the lifted latents overtake the target in its site’s Top-K.

The diagnostic reads the whole code at the target’s site, every latent’s pre-activation and the SAE’s Top-K set at the anchor, in the unmodified model, the empty circuit, and the fitted circuit under each ablation method, on the held-out contexts. It reports the target’s rank and Top-K membership; the number of latents lifted above the unmodified model’s Top-K cut, against the empty circuit; the overlap of the circuit’s Top-K set with the unmodified one; and the faithfulness of *siblings*, latents in the unmodified Top-K at the anchor on at least half of the training contexts, computed exactly as the target’s, against a control group of the site’s other active latents. A circuit whose siblings are restored as well as its target is flagged as an amplifier.

In a pilot, most circuits are specific: the target is restored near 1 while siblings and other active latents are restored far less. Under mean ablation, however, circuits lift many latents above the cut that the empty circuit does not, a minority of circuits are amplifiers that pass every faithfulness score, and several deep attention targets lose their Top-K place to lifted rivals although their pre-activation is reproduced. The rank-keep term (Section 4.3) keeps the target in its Top-K more often but does not reliably remove amplifiers, so specificity is reported for every circuit rather than claimed for the method. The production run confirms the pilot at scale (15,004 circuits): under Z, A, and C the target is restored to a median 0.94, 0.99, and 0.92 of its pre-activation, its siblings to 0.26, 0.46, and 0.41, and the site’s other active latents to 0.14, 0.28, and 0.24. Under the two mean ablations, circuits lift a median of 578 and 517 latents above the unmodified Top-K cut, against 5 and 4 for the empty circuit, and the target stays in its Top-K on 84% and 80% of held-out anchors. 30% of circuits are flagged as amplifiers under at least one ablation method (44% of MLP, 30% of residual-stream, and 11% of attention-output targets), including 38% of passing circuits; 29% of all circuits pass and carry no flag.

The loss terms we tried for rank and specificity before adopting rank-keep are in Appendix O.1.

I Extended Related Work

Position-aware circuits. Position handling divides prior work into three strategies, each right for its setting: position-indexed nodes on templatic data (Marks et al., 2025), per-instance graphs built one prompt at a time (Ameisen et al., 2025), and schema-aligned positions across variable-length examples (Haklay et al., 2025). Our node sets keep the position axis with none of these commitments, using per-position selection unioned over each sequence’s causal prefix, for the setting none of the three targets: one reusable circuit over arbitrary non-templatic text (Appendix C.3).

Contrast contexts. Retrieved non-activating examples appear in automated interpretability for explanation scoring (Paulo et al., 2024); Paulo and Belrose (2025) find evaluations collapse when negatives come from neighbouring features, convergent with our random-beats-close result (Appendix G); and O’Neill and Bui (2024) design corrupted negatives for task-level circuit identification. We use retrieved contrast contexts for causal discovery rather than explanation scoring, and treat their construction as a controlled variable with measured consequences.

The SAE basis. Static structure among latents is well documented: co-occurrence and decoder geometry reveal modularity (Li et al., 2025), coactivation-derived components steer behaviour (Deng et al., 2026), meta-decompositions show latents are not atomic (Leask et al., 2025), cross-layer feature-flow graphs track latents through the forward pass (Laptev et al., 2025), though decoder similarity misses most causally strong feature interactions (Droste et al., 2026), and Jacobian SAEs train adjacent dictionaries for sparse computation between them (Farnik et al., 2025). But that structure is correlational, geometric, or weight-based, where the relationships discovered here are context-conditioned and intervention-tested (Appendix F). The basis itself is imperfect: reconstruction errors are pathological (Gurnee, 2024), downstream utility is mixed (Smith et al., 2025), and units are not canonical (Leask et al., 2025). This conditions our claims: reconstruction residuals stay as explicit anchors, evaluation intervenes on measured activations, and the SAE basis is treated as a useful but imperfect sample of the model’s internal variables.

J Extended Discussion

The non-identifiability probes. The first five probes below were measured on unweighted circuit masking before scaling coefficients were added, the last two on WCM (Section 5.5). A float32 and a bfloat16 run differ only in rounding, yet exchange 16.5% of nodes at layer 8 (Jaccard 0.835) at unchanged faithfulness and size. Raising the sparsity price swaps nodes rather than nesting them (containment 0.49–0.80 between consecutive steps); no gate ever freezes, so the stopping step is another sampling axis; 8–25% of nodes sit within 0.05 of the decision cut and are load-bearing; and short-budget and long-budget runs at matched size share only 0.40–0.53 of their nodes. Recomputing the mean-ablation values without a quarter of their contexts, a change that leaves the pass rate and median faithfulness unchanged, moves node sets as far as a change of hardware (median Jaccard 0.66). A pre-v1 running example’s full-budget and quarter-budget circuits both reconstruct the target, yet at 192 and 250 nodes they overlap at only Jaccard 0.60.

Report node sets as equivalence classes. The non-identifiability results (Section 5.5) change what a responsible node-set claim looks like. Set-level and functional claims (size, faithfulness, the cost findings) are stable under every perturbation we applied. Per-latent claims (“latent g belongs to target f ’s circuit”) are not, since node sets for the same target share only 40–85% of their latents: the objective admits an equivalence class of near-equivalent optima, and a single run samples one of them. Claims of the first kind carry this paper; claims of the second kind require survival across perturbations (precision, sparsity path, stopping step) before they are asserted, and the same redundancy that frustrates them is itself a finding: many latents each doing a small share of the same job is exactly what the collective-faithfulness picture predicts.

From circuits to chains. The co-activation results close a loop in the premise: if latent dependencies were recoverable from co-occurrence, latent-targeted circuit discovery could be served by cheap statistics, and they are not. A weighted circuit is one hop: a set of upstream latents that reproduces its target alone, not yet a multi-hop pathway. Because every circuit is indexed by its target, composition is available in principle: splice the circuit that explains an upstream latent into the circuits that use it, then re-validate by intervening at the earliest nodes and measuring propagation through intermediates. Non-identifiability sets its boundary condition: each link is one sample from an equivalence class, so composition must manage classes of node sets, not single ones. That composition-with-validation step, rather than more discovery methods, is the path from dependency neighbourhoods to genuine circuits.

K Pseudocode for the Gradient Methods

Algorithms 1, 2 and 3 summarise the discovery procedures of Appendix C.1 as implemented; Algorithms 4 and 5 summarise restoration-mode attribution (Appendix C.2) and weighted circuit masking (Section 4.3). Thresholds $(\tau_{\text{act}}, \tau_{\text{neg}}, \tau_{\text{sup}}, \tau_{\text{ind}}, \tau_{\text{prune}})$ and top- K sizes are configuration constants shared across the grid; upstream latents additionally require a minimum corpus active count n_{min} , which filters dead or near-dead latents. All forward passes run with identity passthroughs injected at every SAE site (Appendix C.1), so gradients flow to upstream latent activations, and the target-intervention scores ϕ_{ind} and ϕ_{sup} are computed exactly as in Section 4.4. Hybrid fusion (Algorithm 3) unions the two circuits by latent identity, $\mathcal{F}^{\text{hyb}} = \mathcal{F}^{\text{ctr}} \cup \mathcal{F}^{\text{ab}}$, each node keeping its per-source scores and each directed edge inheriting the higher attribution; optional leave-one-out pruning drops nodes with negligible effect on either acceptance metric, and acceptance requires the fused circuit to clear the sufficiency-to-induce and/or necessity thresholds, with the conjunction set per experiment.

Algorithm 1 Contrastive gradient discovery (one target f)

Require: target f ; activating contexts $\mathcal{P}_f$; contrast-context source $m \in \{\text{close, random, distant}\}$

- 1: $\mathcal{N}_f \leftarrow \text{CONTRASTCONTEXTS}(f, m)$ $\triangleright$ retrieved store, random corpus sequences, or maximally dissimilar sequences
- 2: $\mu_f^+ \leftarrow \text{mean target activation over } \mathcal{P}_f$
- 3: $(\tau_{\text{act}}, \tau_{\text{neg}}) \leftarrow (\tau_{\text{act}}, \tau_{\text{neg}}) \cdot \max(\mu_f^+, 0.1)$ $\triangleright$ scale thresholds so focal targets are not penalised
- 4: **for** each contrast context $x \in \mathcal{N}_f$ **do**
- 5: forward pass; capture pre-activation $\hat{a}_f(x, t)$ at the target's site
- 6: $t^* \leftarrow \arg \max_t \hat{a}_f(x, t)$; $\mathcal{L}^{\text{ctr}}(x) \leftarrow (\hat{a}_f(x, t^*) - \mu_f^+)^2$
- 7: backward pass: $s_g^{\text{ctr}}(x) \leftarrow a_g(x, t_g) \cdot \partial \mathcal{L}^{\text{ctr}}(x) / \partial a_g(x, t_g)$ for every upstream latent g
- 8: **end for**
- 9: $S_g^{\text{ctr}} \leftarrow \frac{1}{|\mathcal{N}_f|} \sum_x s_g^{\text{ctr}}(x)$ for every g
- 10: $P \leftarrow \text{top-}K \text{ latents by } S_g^{\text{ctr}} \text{ with } S_g^{\text{ctr}} \geq \tau_{\text{act}} \text{ and active count } \geq n_{\text{min}}$
- 11: $N \leftarrow \text{top-}K \text{ latents by } -S_g^{\text{ctr}} \text{ with } |S_g^{\text{ctr}}| \geq \tau_{\text{neg}} \text{ and active count } \geq n_{\text{min}}$
- 12: $C \leftarrow \text{star circuit } \{g \rightarrow f\} \text{ over } P \cup N, \text{ edge weights } S_g^{\text{ctr}}$
- 13: **if** $\tau_{\text{prune}} > 0$ **then**
- 14: leave-one-out prune nodes whose removal changes ϕ_{ind} by less than τ_{prune}
- 15: **end if**
- 16: $(\phi_{\text{ind}}, \phi_{\text{sup}}) \leftarrow \text{target-intervention evaluation of } C \text{ on } \mathcal{P}_f \text{ and } \mathcal{N}_f$
- 17: **return** C if $\phi_{\text{ind}} \geq \tau_{\text{ind}}$, otherwise reject

Algorithm 2 Ablation gradient discovery (one target f)

Require: target f ; activating contexts $\mathcal{P}_f$

- 1: **for** each activating context $(x, t) \in \mathcal{P}_f$ **do**
 - 2: forward pass; backward pass from the target’s sparse activation $a_f(x, t)$
 - 3: $s_g^{\text{ab}}(x) \leftarrow a_g(x, t_g) \cdot \partial a_f(x, t) / \partial a_g(x, t_g)$ for every upstream latent g
 - 4: **end for**
 - 5: $S_g^{\text{ab}} \leftarrow \frac{1}{|\mathcal{P}_f|} \sum_x s_g^{\text{ab}}(x)$ for every g
 - 6: $P \leftarrow$ top- K latents by S_g^{ab} above threshold, active count $\geq n_{\min}$
 - 7: $C \leftarrow$ star circuit $\{g \rightarrow f\}$ over P , edge weights S_g^{ab}
 - 8: **if** $\tau_{\text{prune}} > 0$ **then**
 - 9: leave-one-out prune nodes whose removal changes ϕ_{sup} by less than τ_{prune}
 - 10: **end if**
 - 11: $(\phi_{\text{ind}}, \phi_{\text{sup}}) \leftarrow$ target-intervention evaluation of C
 - 12: **return** C if $\phi_{\text{sup}} \geq \tau_{\text{sup}}$, otherwise reject
-

Algorithm 3 Hybrid gradient discovery (one target f)

Require: target f ; acceptance mode $\in \{\phi_{\text{ind}}, \phi_{\text{sup}}, \text{both}, \text{either}\}$

- 1: $C_{\text{ctr}} \leftarrow$ Algorithm 1 on f ; $C_{\text{ab}} \leftarrow$ Algorithm 2 on f ▷ either may reject
 - 2: **if** both rejected **then**
 - 3: **return** reject
 - 4: **end if**
 - 5: $C \leftarrow$ fuse returned circuits by latent identity: union of nodes, each retaining its per-source attribution scores; each directed edge keeps the higher attribution
 - 6: **if** $\tau_{\text{prune}} > 0$ **then**
 - 7: leave-one-out prune nodes with negligible effect on either acceptance metric
 - 8: **end if**
 - 9: $(\phi_{\text{ind}}, \phi_{\text{sup}}) \leftarrow$ target-intervention evaluation of the fused C
 - 10: **return** C if the acceptance mode’s condition holds ($\phi_{\text{ind}} \geq \tau_{\text{ind}}$ and/or $\phi_{\text{sup}} \geq \tau_{\text{sup}}$), otherwise reject
-

Algorithm 4 Restoration-mode attribution (one target f ; `ig_restoration` substitutes per-round integrated-gradient scoring in line 4)

Require: target f ; activating contexts $\mathcal{P}_f$; round budget $R =$ number of upstream sites; per-round admission size k_r

- 1: $M \leftarrow \emptyset$; ablate every upstream latent (every latent at its ablation value)
 - 2: **for** round $r = 1, \dots, R$ **do**
 - 3: forward with nodes M recomputed from the modified stream and all other latents ablated; capture the target pre-activation $\hat{a}_f$
 - 4: score every latent $g \notin M$ by gradient-times-activation at the *current* partially restored stream
 ▷ re-linearisation: later rounds see the consequences of earlier admissions
 - 5: $M \leftarrow M \cup \{\text{top-}k_r \text{ latents not in } M \text{ by score}\}$
 - 6: **if** $\hat{a}_f$ is within tolerance of its natural value **then break** ▷ restoration certificate
 - 7: **end if**
 - 8: **end for**
 - 9: $C \leftarrow$ circuit over M with per-round scores; target-intervention and ablation-faithfulness evaluation as configured
 - 10: **return** C
-

Algorithm 5 Weighted circuit masking, WCM[Z+C+A] with rank-keep (one target f ; protocol v1, Table 4)

Require: target f ; activating contexts $\mathcal{P}_f$ and contrast contexts $\mathcal{N}_f$, each split 48 train / 16 held out;
 $\lambda = 10^{-3}$; $\gamma_C = \gamma_A = 0.25$; $\gamma_R = 3 \times 10^{-3}$; steps $S = 400$

- 1: target $\leftarrow$ natural pre-activation of f at the training anchors (no-grad forward); $\nu \leftarrow \text{mean}(\text{target}^2)$
- 2: $r \leftarrow$ rank of f among its site's other latents at each training anchor, unmodified model
- 3: ablation values: $C \leftarrow$ site means over the *training* contrast contexts; $A \leftarrow$ over the *training* activating contexts
- 4: initialise $\theta_s = 4$, $\psi_s = \text{softplus}^{-1}(1)$ for every upstream site s
- 5: **for** step = $1, \dots, S$ **do**
- 6: $T \leftarrow 0.05^{\text{step}/S}$; gates $m_s \leftarrow \sigma(\theta_s/T)$; coefficients $\alpha_s \leftarrow \text{softplus}(\psi_s)$ $\triangleright$ geometric anneal $1 \rightarrow 0.05$
- 7: **for** $\pi \in \{Z, C, A\}$ **do** $\triangleright$ same θ, ψ under all three ablation methods
- 8: forward with kept latents at $\alpha \times$ live value, closed latents at their π ablation value
- 9: $\mathcal{L}_\pi \leftarrow$ MSE of f 's pre-activation against target; $\mathcal{R}_\pi \leftarrow \text{mean rel}(\tau_\pi^{(r)} - \hat{a}_f)^2$
- 10: **end for**
- 11: AdamW step on $\sum_\pi \gamma_\pi (\mathcal{L}_\pi + \gamma_R \mathcal{R}_\pi) / \nu + \lambda \sum_{s,i} [\sigma(\theta_{s,i}) + (1 - m_{s,i}) |\alpha_{s,i} - 1|]$
- 12: **end for**
- 13: circuit $\leftarrow \{(i, \alpha_i) : m_i > \frac{1}{2}\}$; evaluate on the held-out contexts under Section 4.4
- 14: **return** circuit

Setting	Value
<i>Models and dictionaries</i>	
TuringLLM-1.0-254M	Top-K, $k = 128$, 40,960 latents; attention output, MLP output, residual stream at 12 layers
Pythia-70m	EleutherAI Top-K, $k = 16$, 32,768 latents; the same three sites at 6 layers
GPT-2 small	OpenAI Top-K, $k = 32$, 32,768 latents, layer-normalised input; attention output, residual stream between attention and MLP, MLP output, residual stream at 12 layers
Targets per model	TuringLLM: 15,046, sampled across every site with an upstream SAE site (layer-0 attention has none); <i>[pending: Pythia-70m and GPT-2 small, one sampling frame, DAN-9]</i>
<i>Contexts</i>	
Activating contexts	64 strongest and 64 mid-band stored contexts (Section 4.1); train on 32 strongest and 16 mid-band; hold out 16 strongest (primary) and 16 mid-band (reported). Stratified split: contexts ranked by anchor activation in blocks of 4, block b holds out position $(b + 2) \bmod 4$, so ranks 1–2 always train. Targets with no mid-band store train on 48 strongest (665 of 15,004)
Contrast contexts	64 per target: corpus sequences ranked by highest cosine similarity to any activating context (pooled final-layer representations), activating contexts excluded, kept only if the target is outside its Top-K (Appendix L); 48 train, 16 held out, stratified by similarity
Anchor	position of the target’s peak activation in each context
<i>Weighted circuit masking, WCM[Z+C+A]</i>	
Ablation methods	Z zero ablation (weight 1); C mean over contrast contexts ($\gamma_C = 0.25$); A mean over activating contexts ($\gamma_A = 0.25$), at every layer; both means computed on the 48 training contexts only
Objective	MSE to the target’s natural pre-activation at the anchors, each term divided by one shared $v = \text{mean}(\text{target}^2)$
Rank-keep	$\gamma_R = 3 \times 10^{-3}$ on $\text{relu}(\tau^{(r)} - \hat{a}_f)^2$ inside every ablation term, r the target’s rank in the unmodified model at the anchor
Gates	$m = \sigma(\theta/T)$, $\theta_0 = 4$; T annealed geometrically from 1 to 0.05; nodes are $\{m > \frac{1}{2}\}$
Scaling coefficients	$\alpha = \text{softplus}(\psi)$, $\alpha_0 = 1$; closed gates charged $\lambda(1 - m) \alpha - 1 $
Sparsity price	$\lambda = 10^{-3}$, flat per latent
Optimiser	AdamW, learning rate 0.05 (constant), weight decay 0.05, 400 steps, batch 4
Execution	nodes recomputed from the edited stream; SAE errors preserved; every upstream site edited
<i>Evaluation (Section 4.4), on the 16 held-out contexts</i>	
Read	every pass/fail score reads the target’s activation (after Top-K), in every term of the ratio; the same scores on the pre-activation are reported as a diagnostic; training uses the pre-activation, which keeps a gradient where Top-K censors the activation
Faithfulness	ϕ_π^{free} under Z, dense A, sparsity-preserving A and C
Necessity	$\phi_{\text{sup}}(C; \pi) = (a_{\text{pos}} - a_{M \setminus C}) / (a_{\text{pos}} - a_\emptyset)$, the circuit as one set
Sufficiency to induce	ϕ_{ind} and ϕ_{ind}^α on contrast contexts
Clamped faithfulness	ϕ_π^{pin} under the same ablation methods
Random-circuit baseline	size- and site-matched, drawn from live latents, coefficients fitted with the same budget; <i>[pending: number of draws, DAN-46]</i>
Specificity (reported)	the target’s rank and Top-K share; latents lifted above the unmodified Top-K cut, against the empty circuit; sibling faithfulness against the target’s (Appendix H)
Near-threshold targets	median clean rank at the anchor $\geq K/2$, identified from the unmodified model; kept in the headline pass rate and also reported as a stratum on both reads
Pass criterion	faithfulness under Z, sparsity-preserving A, and C all in $[0.8, 1.5]$, and necessity ≥ 0.9 (held-out strongest contexts, activation read); vacuous denominators excluded and counted; the largest deviation $\max_\pi \phi_\pi^{\text{free}} - 1 $ reported alongside

Table 4: The protocol behind every reported circuit. Values marked pending are fixed before the corresponding runs are reported.

L Context Stores and Contrast Contexts

L.1 Context Store Formalism

The collection pass assigns each latent a scored context set $C_f = \{(x, t), a_f(x, t) : a_f(x, t) > 0\}$. The top-context store retains the K_{top} highest-scoring active contexts,

$$\text{TopCtx}_K(f) = \arg \top_{S \subseteq C_f, |S|=K_{\text{top}}} \sum_{((x,t),s) \in S} s,$$

approximating the upper tail of the conditional activation distribution. The mid-context store is a bounded reservoir sample from the sigma band,

$$\text{MidCtx}_M(f) \sim \text{ReservoirSample}_M(\{(x, t) : \mu_f + 0.5 \sigma_f \leq a_f(x, t) \leq \mu_f + 1.5 \sigma_f\}),$$

maintained without sorting every active occurrence. Mid contexts distinguish a latent’s ordinary operating regime from its most vivid examples and provide probe data where extreme contexts would be too narrow.

Contrast-context retrieval. Let $r(x) \in \mathbb{R}^{1024}$ be the retrieval representation of sequence x : the final layer’s residual stream, mean-pooled over positions, captured once during collection, and let $\mathcal{P}_f$ be the sequences stored as activating contexts for f . The query is the normalised centroid of the activating contexts

$$c_f = \text{normalize} \left(\frac{1}{|\mathcal{P}_f|} \sum_{x_p \in \mathcal{P}_f} r(x_p) \right),$$

and the store retains the $n = 64$ sequences most similar to c_f that are not stored activating contexts:

$$\text{ContrastCtx}_n(f) = \arg \top_{S \subseteq \mathcal{D} \setminus \mathcal{P}_f, |S|=n} \sum_{x \in S} \langle c_f, r(x) \rangle.$$

Retrieval is exact rather than approximate: representations are ℓ_2 -normalised and full cosine similarities are computed by chunked matrix multiplication on GPU, querying the $K = 512$ nearest candidates before the filter that removes stored activating contexts. This store serves the attribution baselines. Its representation index is a sample of 2×10^5 sequences, so for targets none of whose activating contexts fall in the sample the centroid is undefined, and the store returns one shared default set for all of them (38% of TuringLLM targets); the rest are anchored on the few activating contexts the sample contains.

Contrast contexts for WCM. Weighted circuit masking and its evaluation instead select contrast contexts per target at fit time. Every activating context is represented, read from the index when present and otherwise computed by a forward pass; candidates from the sampled corpus are ranked by their highest cosine similarity to any activating context, $\max_{x_p \in \mathcal{P}_f} \langle r(x_p), r(x) \rangle$, with the activating contexts excluded; and each candidate is run through the model and kept only if the target is outside its site’s Top-K at every position, until 64 are selected. The silence check rejects between 14% and 51% of the closest candidates on the targets we inspected, so unchecked retrieval would place many contrast contexts where the target fires. The same selector with random or maximally dissimilar ranking gives the *random* and *distant* sources of Appendix G.

Logit faithfulness. The output-level score of Section 5.4 is the faithfulness formula of Section 4.4 with the target’s value replaced by the output metric m , the mean log-probability of the actual next token at the anchor: $(m(C) - m(\emptyset)) / (m(M) - m(\emptyset))$, with every in-scope latent outside the circuit mean-ablated.

M Evaluation Protocol Studies

M.1 Why Position Aggregation Fails: Union Coverage

The SAE code is k -sparse per position, so a position-collapsed circuit evaluated across a target’s causal prefix must cover the *union* of every position’s active set. Measured on a target’s probe batch, that union is large at every site (roughly 1–3.7k distinct latents per attention-output site, 7–10k per residual-stream site, and 7–11.8k per MLP-output site), and the burden concentrates in the additive residual carrier: mean-ablating any single residual-stream site alone sends the target to 0.000, while ablating any single attention-output site leaves 0.90–0.96. The harness passes its identity check, keep-all identity at the reduced-precision noise level (0.994), and a selection-agnostic density sweep reproduces the wall; two scoring defects found later in this harness (a mixed pre-/post-Top-K read and ablation limited to member sites) are corrected in protocol v1 (Appendix B), so the numbers here are indicative; position-collapsed faithfulness failure is a property of the dense residual stream plus position aggregation, not an artifact of the harness.

M.2 Sparsity-Preserving Mean Ablation

The classic dense mean ablation assigns every latent outside the circuit its mean, producing a stream with tens of thousands of simultaneously active latents per position, outside the state space a Top-K SAE can produce. The sparsity-preserving variant applies the means and then re-imposes the per-position Top-K (kept latents, then the largest means of the other latents up to k , rest zero), which restores the keep-all identity to exactly 1.0. At scale (32 targets, paired) the sparsity-preserving variant does not systematically shift faithfulness: it raises both a_C and $a_\emptyset$, leaving the ratio a wash (50W/46L). It therefore functions as a cleaner protocol rather than a score inflator; but at small node sets and tight Top-K budgets the two variants can disagree on *rankings*, which is why the body’s comparisons name their ablation method.

M.3 Sources of the Mean Ablation

The empty-circuit baseline $a_\emptyset$ is a measured object with its own failure modes. Means over *activating* contexts leak target signal into $a_\emptyset$ increasingly with depth ($a_\emptyset/a_{\text{pos}} = 0/23/30/26\%$ at layers 2/8/9/10); means over the target’s *contrast* contexts hold $a_\emptyset = 0.0000$ on every target measured, even under the hardest close contrast contexts. As an ablation method during *discovery*, however, contrast-context mean ablation is neutral for path attribution and destructive for restoration (recomputed faithfulness -0.07 to -0.98 , +26% nodes, monotone in contrast-context distance), another instance of discovery and evaluation ablation methods needing to be chosen separately and stated. A three-way cross-tab of mean sources (the target’s activating contexts, a random corpus sample, and a farthest-point sample of the corpus by sequence representation) shows method *rankings* are robust to the source while absolute levels shift with how much target signal the mean carries, and sufficiency to induce is invariant to it; every reported number therefore names its ablation method.

N Attribution-Mode and Pruning Studies

Comparisons in this appendix use within-target pairing or depth stratification throughout.

N.1 Recurrence Pruning

A large tail of nodes appears in exactly one probe sequence: context-specific scaffolding rather than mechanism. Pruning nodes that fire in fewer than two probe sequences removes $\sim 10\%$ of nodes for ~ 0.010 recomputed faithfulness at every depth; the harsher eight-sequence bar is smaller but fails at depth (recomputed faithfulness 0.680 vs 1.049 at layer 10, and the dense mean-ablation score goes negative at layer 8), so the two-sequence bar is the smallest setting that survives both ablation methods and depth. Nodes from the contrastive gradient are judged on contrast contexts and nodes from the

ablation gradient on activating contexts, and any node the prune cannot observe is exempted rather than dropped.

N.2 Effect Thresholds Cannot Replace Validated Stopping

Holding the ranking fixed and swapping the stopping rule isolates what does the work. A fixed absolute effect threshold at the field-default value sits $\sim 2.3\times$ above our *maximum* node score (it deletes everything), and even percentile cuts calibrated to our own score distribution crater function (the dose-response of Appendix D), while bisection over the identical ranking preserves it. The stopping rule, not the ranking, carries the validity.

N.3 Selection Rules and the Pruning Frontier

A per-target percentile cut on $|\text{attribution}|$ dominates top- N and every per-position-normalised selection rule at equal or better faithfulness with $1.4\text{--}2.7\times$ fewer latents (the driving signal is peaked in a *global* magnitude sense; per-position normalisation wastes budget and needs a dead-position guard). The magnitude-bisection prune then composes with it: percentile selection raises the achievable faithfulness ceiling, the prune lowers size at fixed faithfulness, and the composition dominates on all six test targets (layers 3–11). On the mid-depth residual-stream target, top- N selection cannot reach the requested faithfulness at all, while the composed recipe reaches it at less than half the node count.

N.4 Direct-Effect Edges

Edges are attached with the direct-effect construction of Marks et al. (2025): an edge’s weight is the indirect effect of an upstream node on the metric via its *direct* effect on the downstream node, with intermediate nodes stop-gradiented, computed as batched vector–Jacobian products. Edges never change the node set or any evaluation score; they are descriptive. But their statistics mirror the discovery mode: local attribution yields sparse shallow graphs (~ 9 edges per node, depth ~ 4) where path attribution yields dense deep ones (~ 40 edges per node, depth ~ 10), structure following the estimator’s linearisation. For weighted circuits the same construction is linearised around the circuit’s own run (nodes at their scaled values, every other latent at the ablation value), so the coefficients enter both the upstream change and the downstream Jacobian, and edges are computed separately under each ablation method; a node’s out-edges plus its direct edge to the target then sum exactly to its own attribution, which the implementation reproduces to within 1%. The edges also hold beyond first order: with every other node pinned at its circuit value and the upstream node set to its ablation value, the measured change in the downstream node reproduces the edge weight. In a pilot, the wiring is multi-step rather than a star, with most of a circuit’s attribution reaching the target through other circuit nodes. It is also dense, with positive and negative routes largely cancelling and no small set of edges carrying most of the weight, and edge sets computed under different ablation methods overlap only partly. We therefore interpret only edges that clear a threshold with one sign under all three ablation methods, and state the share of edge weight they carry; *[pending: wiring statistics on the protocol-v1 circuits, DAN-75]*. An edge’s sign combines whether the upstream node raises the downstream one with the downstream node’s marginal effect on the target, so a negative edge need not mean suppression. The residual-stream relay of Section 5.4 is such a shared structure.

N.5 Error Nodes for a Target Latent

Keeping SAE errors as ablatable nodes reproduces, for target latents, the error-node picture reported for output targets (Marks et al., 2025), with one asymmetry in our favour: the empty-circuit anchor is error-free here ($a_\emptyset$ identical with and without error terms, 0.000 at layers 2 and 9), so the collapse of the empty-circuit baseline does not arise. Faithfulness at depth, however, is error-carried: zeroing all upstream error terms leaves recomputed faithfulness ≈ 1.08 at layer 2 but 0.005 at layer 9. Because raw and pruned node sets occupy every upstream site, including error nodes selectively awaits discovery-side error attribution.

O Weighted Circuit Masking Design Ablations

Rejected or superseded designs for WCM, recorded because each closes a natural reviewer question.

O.1 Loss Terms for Target Rank and Specificity

An *off-target* term charging every other site latent for rising above its unmodified and empty-circuit values cut lifts three- to sevenfold at no faithfulness cost, but addressed a property the method need not guarantee (Appendix H) and grew circuits when it also charged harmless lifts below the cut. A *Top-K hinge* requiring only that the target beat the k -th other latent restored faithfulness but, used alone, let circuits keep the target in the Top-K by lifting it together with its neighbourhood. A convex pull towards rank 1 inflated circuits by an order of magnitude at every weight tried, since it never switches off and dominates once the reproduction error is small. The *firing-margin* objective, fitting the target’s lead over the Top-K cut, needs its loss normalised by the target’s pre-activation scale rather than the margin’s, which is near zero for targets at the cut. Rank-keep, which switches off once the target holds its natural rank, is the single term that kept the gains without these costs, and its effect is flat across a thirtyfold range of weights.

O.2 Sparsity Pricing and Normalisation

Five rejected variants of the sparsity price. Gradient-scaled sparsity pricing is *anti*-correlated with need; pricing inversely to site count is non-monotone; predicting the price from target-latent features misses requested sizes by 24–29% where one-probe calibration lands within ~4%; initialising gates from observed activity buys no speedup (the optimiser updates all parameters in parallel, so deadweight costs nothing); and gradient-weighted per-site pricing is dominated by flat pricing, with the methodological footnote that quantiles over mostly-zero gradient tensors measure sparsity, not scale.

Two further design records support Section 4.3. The normaliser for training under two ablation methods is target-scaled and bounded by construction because its predecessor was not: normalising each term by its own fully-closed-mask loss, which is unbounded far from the data, silently annihilated the zero-ablation term at depth (the two closed-mask losses differ there by up to eight orders of magnitude), reducing WCM[Z+C] training to the C term alone; the design rule is that a normaliser must never be able to annihilate its own term. The one-probe size calibration is the formula $\lambda = \lambda_0 (n_{\text{probe}}/n_{\text{req}})^{1/q}$: a single reference run at λ_0 measures n_{probe} , landing requested sizes within a few percent where a per-size search (cf. the sparsity-targeting of Bhaskar et al., 2024) would cost several runs per target.

O.3 Binarisation Sweep

The straight-through estimator makes interior gate values forward-irrelevant, so the sparsity penalty drifts nodes toward just-above-threshold: the decision boundary becomes an *attractor*, and per-step node flips grow through training (218 per step at layer 10), a stopping-step lottery. Annealing instead yields the tightest zero-ablation faithfulness in the programme (0.002–0.10 deviation from unity) at slightly worse mean-ablation scores, another home-turf demonstration. Schedule compression does not help: a 200-step run matches the 400-step metrics at ~2× the per-step churn (churn is priced by per-step pressure, not schedule shape); reach-then-hold leaves churn unchanged and inflates size 21–48%; and truncating a 400-step run at 200 yields node sets 57–72% oversized with the gate still half-soft.

Method	Top-155 ϕ_{ind}	Median ϕ_{ind}	Median nodes	Targets accepted (%)
Contrastive gradient	0.82	0.82	385	40
Ablation gradient	0.97	0.59	193	91
Hybrid gradient	1.07	0.69	264	91

Table 5: Descriptive statistics for the 128-target grid (3 methods $\times$ 3 contrast-context sources; 384 runs per method). Median sufficiency to induce (ϕ_{ind}) and node counts are over each method’s accepted circuits, pooled across sources. The top-155 column is the median over each method’s 155 highest-scoring circuits (155 being the smallest per-method accepted count) for an equal-population comparison; values above 1 are intervention overshoot (Section 4.4). Targets accepted is the share of the method’s 384 target–source discovery runs (128 targets $\times$ 3 sources) whose circuit passed that method’s own intervention gate.

Method	Median ϕ_{ind}	Median ϕ_{sup}	Targets accepted (%)
Sparse co-activation expansion	—	—	0
Differential activation	—	—	0
Ablation gradient (reference)	0.71	1.00	95

Table 6: Correlational discovery baselines under the same 128-target protocol, evidence base, and target-intervention gates as the gradient methods (Appendix F). Dashes indicate that no circuit passed the gates, so no accepted-circuit medians exist. The ablation gradient row is the same-batch reference (121 of 128 targets accepted).

P Additional Results

P.1 The Attribution Grid

The *grid*: 128 component-balanced random target latents, each processed by the contrastive, ablation, and hybrid attribution arms (Appendix C) under all three contrast-context sources, for 1,152 discovery runs, summarised in Table 5. Acceptance gates differ by method (Appendix K), so acceptance rates are not comparable across methods. The *panels* behind Appendix D use three to sixteen depth-spanning targets per experiment (layers 2–11, all component kinds); they are directional measurements. Contrastive gradient separates discovery inputs from evaluation inputs by construction; ablation gradient attributes and evaluates on the same probe set.

P.2 Co-Activation with Fitted Scaling Coefficients: the Strongest Correlational Baseline

Appendix F shows correlational proposals failing the intervention gates outright; the fairest possible strengthening of that challenger is to hand it the weighted-circuit machinery itself. Take the n latents with the largest summed activation at the target’s anchors (size-matched to the discovered weighted circuit), and score them twice through the identical pipeline: at natural values ($\alpha = 1$), and with scaling coefficients fitted by the same objective, ablation methods, and budget as the discovered run.

At natural values the selection fails everywhere, and its failure is directional: 0.08–0.51 at layer 2, but 7.6–9.1 at layer 9 — co-activation picks latents that fire alongside the target and misses the rest of the stream that normally holds it in range, so keeping only its picks *overdrives* the target several-fold. With fitted coefficients it becomes the strongest baseline in the programme: zero-ablation faithfulness reaches ~ 1 at layer 2 on every target. The joint suite still separates it from the discovered circuits. Under the production recipe, WCM[Z+C+A], best against best, the discovered weighted circuits hold faithfulness under both zero and mean ablation on all five targets (three at layer 2, two at layer 9) (zero ablation 0.92–1.06, mean ablation 0.86–1.18), while the co-activation sets hold both on none: at layer 2 the fitted coefficients calibrate against one background and overshoot the other (mean ablation 1.30–2.47, coefficients reaching 6–10 $\times$), and at layer 9 no coefficient assignment reconciles

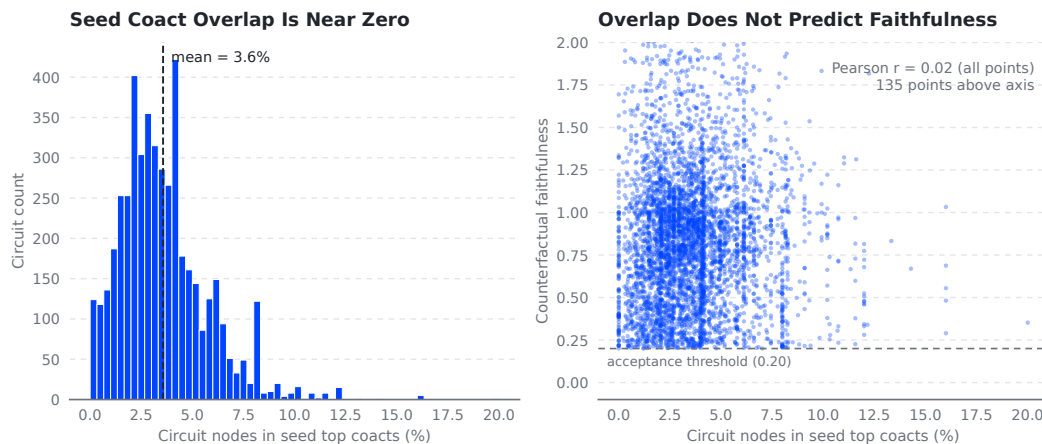


Figure 14: Discovered dependencies are not co-activation neighbourhoods (Appendix F). Left: distribution of the percentage of each circuit’s nodes that appear among its target’s top co-activation partners (mean 3.6%). Right: the same overlap against the circuit’s sufficiency to induce (Pearson $r = 0.02$ over all points; 135 of 4,741 points lie above the truncated axis). The dashed line marks the acceptance threshold (0.20), the minimum sufficiency to induce an accepted circuit can have. Statistics are over the 4,741 circuits with post-analysis co-activation metrics from the 7,525-circuit run.

the co-activation selection with all three ablation methods at once (zero ablation 3.09 on one target, 0.79 on the other). These are TuringLLM targets under the pre-v1 recipe (*[pending: rerun under protocol v1]*). Reconstruction alone is therefore spoofable by a calibrated correlational set, and the revision to Appendix F is precise: correlational *selection* without coefficients fails outright, and with coefficients it fails calibration — what it cannot buy at any coefficient setting is a node set whose contribution is invariant to the ablation method, which is what weighted circuit masking selects for.